

Chapter 1

INTRODUCTION

The rapid growth in medical imaging volume, increasing patient load, and fragmentation of hospital information systems have created significant challenges in modern healthcare delivery. Radiological imaging, laboratory reports, and clinical history are often stored in separate platforms such as PACS (Picture Archiving and Communication System), LIS (Laboratory Information System), and EHR (Electronic Health Record). Clinical interpretation therefore requires manual cross-referencing by physicians, which increases cognitive workload and may lead to delays or inconsistencies in diagnosis.

Recent advances in Artificial Intelligence, particularly in Computer Vision and Natural Language Processing, have demonstrated strong potential in automating radiology report generation and medical image interpretation. However, most existing systems focus primarily on single-modality analysis, typically converting images into reports, without performing correlated reasoning across imaging, biomarkers, and clinical metadata

ScanWise AI is proposed as a Multimodal Clinical Intelligence System that integrates medical imaging, laboratory reports, and transplant allocation logic into a unified decision support framework. The system aims to reduce diagnostic inconsistencies, improve triage efficiency, and support high-risk decisions such as organ transplantation.

1.1 Background

Radiology has become a cornerstone of modern diagnostic medicine. Imaging technologies such as CT and MRI provide high-resolution visualization of anatomical structures and pathological changes, enabling early detection and monitoring of diseases ranging from infections to malignancies. Over recent years, artificial intelligence has increasingly been integrated into radiological workflows. Deep learning models, particularly Convolutional Neural Networks, have demonstrated high performance in disease classification tasks. Transformer architectures and Vision-Language Models have further advanced the field by enabling automated radiology report generation and visual question answering systems. These developments reflect a broader shift toward AI-assisted clinical interpretation.

Despite this progress, existing systems predominantly focus on the conversion of images into structured reports or classification labels. Research surveys between 2021 and 2025 indicate that most Radiology Report Generation frameworks prioritize linguistic quality metrics and image-to-text alignment rather than multimodal clinical reasoning.

While these approaches reduce documentation workload, they rarely extend beyond image-based interpretation. In practical hospital settings, diagnosis is rarely isolated to imaging findings. Physicians correlate scan abnormalities with laboratory biomarkers such as C-Reactive Protein, White Blood Cell count, and hemoglobin levels, along with patient history and comorbidities.

Another critical domain is organ transplantation. Allocation decisions depend on standardized scoring systems such as MELD for liver disease and LAS for lung disease. Compatibility parameters including ABO blood group matching and Human Leukocyte Antigen compatibility are evaluated alongside Cold Ischemia Time, which measures the duration an organ remains preserved outside the body. Current transplant management systems calculate these parameters but do not dynamically integrate radiological severity assessment or biomarker-based infection risk into allocation logic.

Therefore, the background context reveals a clear technological gap. Existing AI models excel at single-modality tasks but lack unified cross-domain reasoning capabilities. There remains a need for a scalable system capable of integrating image analysis, text extraction, clinical metadata, and transplant allocation constraints within a single framework. ScanWise AI is positioned to address this gap by combining computer vision, natural language processing, and structured rule-based reasoning to deliver clinically coherent outputs.

1.2 Statistics

The current landscape of medical imaging and diagnostic services reveals significant growth in imaging utilization worldwide alongside notable shortages in specialist manpower, which underscores the need for intelligent support systems such as the one proposed in this project. Globally, medical imaging has become one of the most commonly used diagnostic tools in healthcare, with modalities such as CT (Computed Tomography) and MRI (Magnetic Resonance Imaging) increasingly integrated into routine clinical workflows. Although precise global counts of annual imaging procedures are not centrally published in one source, multiple

industry and medical-device reports from the last few years confirm a steady rise in advanced imaging usage such as CT and MRI scans across regions, reflecting year-over-year expansion in diagnostic volumes and health service demand. This increased utilization is driven by aging populations, expanding disease burden, greater awareness of early detection benefits, and broader availability of imaging modalities in both public and private sectors.

In the Indian context, these global trends are mirrored but complicated by resource limitations. According to recent professional assessments, India faces a significant shortage of radiologists relative to population and imaging workload. Estimates suggest that there are approximately 10,000–15,000 radiologists serving a population of roughly 1.4 billion people, which equates to an extremely high ratio of individuals per radiologist. Some reports indicate a ratio as extreme as one radiologist for every 65,000 people, despite annual increases in imaging volume of 15–20%. Such ratios are far below global standards and point toward increasing workload burdens for each specialist, contributing to longer reporting times and higher risk of fatigue-related errors.

Further illustrating the manpower challenge, additional industry data has reported figures indicating that India has one radiologist per 100,000 individuals when considering broader counts of imaging professionals across the country. This severe imbalance hampers the ability of healthcare systems to meet diagnostic demands, particularly in rural and lower-resource regions of India where access to trained radiologists and advanced imaging services is limited. Beyond manpower shortages, diagnostic imaging equipment availability also reflects systemic constraints. While precise national inventories of CT and MRI equipment for India are not routinely published, government-commissioned reports indicate that the number of MRI units in India remains low relative to population size. A sample industry report from recent years suggested an estimated total of roughly 3,500–4,000 MRI machines in operation, equating to approximately three MRI scanners per million population. This is significantly lower compared to OECD averages and reflects the broader challenges in providing high-throughput imaging services at scale.

Organ transplantation statistics also provide context for the clinical demand that imaging and diagnostic systems must support. Globally, over 130,000 solid organ transplants are performed each year, indicating the high volume and clinical complexity associated with modern transplant programs. In India specifically, approximately 17,000–18,000 solid organ transplants

are conducted annually, positioning the country among the highest in total volume despite lower transplant rates per million population compared to high-income countries. These figures reflect increasing success but also highlight inequities in access, affordability, and clinical resource allocation.

These statistics collectively demonstrate a clear gap between rising imaging and transplantation needs and the availability of radiological expertise and equipment. The growth of imaging volumes, ongoing manpower shortages, and expanding clinical complexity underscore the pressing need for automated assistance systems capable of augmenting diagnostic workflows while maintaining clinical quality and safety.

1.3 Prior Existing Technologies

Recent research in multimodal medical artificial intelligence has increasingly shifted from single-modality image analysis toward integrated vision-language systems capable of generating radiology reports and answering clinical queries. One of the significant advancements in this direction is the development of multimodal frameworks designed specifically for volumetric 3D medical data. In [1], a system is introduced that performs both automated report generation and visual question answering directly from 3D abdominal CT scans. Unlike traditional vision-language models that operate on 2D images such as chest X-rays, this framework incorporates a dedicated 3D encoder pretrained using contrastive alignment between image and text embeddings. The architecture consists of a 3D vision backbone, a projection layer for cross-modal alignment, and a language model fine-tuned for clinical text generation. The study addresses dataset imbalance by refining training sets and rebalancing answer choices, demonstrating improved performance in both report generation and question answering tasks. Evaluation is conducted using the GREEN metric for factual consistency and standard accuracy measures for VQA. Although the approach confirms that volumetric data can be integrated into vision-language models, the reported performance remains moderate, and the authors highlight limitations related to dataset bias, evaluation sensitivity, and computational cost. The study establishes the feasibility of 3D multimodal reasoning but also reveals the complexity of handling volumetric depth and maintaining clinical reliability.

A broader conceptual extension of multimodal AI for 3D medical imaging is discussed in [2], where 3D scans are treated analogously to long video sequences. The central idea is that a CT

or MRI volume can be processed as a sequence of frames along a spatial axis, allowing adaptation of video-text generative models originally designed for natural videos. This work emphasizes structural similarities between temporal video frames and spatial CT slices, proposing that stacking slices enables reuse of large-scale video-language architectures. A key conceptual contribution is the introduction of “self-multimodality,” where multiple imaging phases such as arterial and venous CT phases or different MRI sequences are interpreted jointly to reflect real radiological reasoning. The study further stresses the integration of metadata, scan protocols, and patient context as essential components of multimodal understanding. However, it acknowledges major barriers including scarcity of large paired 3D datasets, privacy concerns in volumetric imaging, and the absence of standardized evaluation benchmarks for cross-sequence reasoning. This perspective shapes the direction of future multimodal systems but does not present a fully validated clinical solution.

Another category of prior technologies focuses on classical multimodal deep learning architectures that fuse image and text features through structured pipelines. In [3], a joint framework is proposed that combines Convolutional Neural Networks for image feature extraction with Bidirectional LSTM networks and attention mechanisms for clinical text understanding. The image branch captures spatial patterns such as lesion boundaries and anatomical relationships, while the text branch encodes semantic and contextual information from reports. A multimodal fusion layer integrates both representations into a shared feature space, enabling tasks such as disease classification, lesion localization, and clinical description generation. Experimental results demonstrate that fused models outperform single-modality baselines in accuracy, recall, F1 score, and localization metrics. Despite strong empirical performance, such systems are typically task-specific and rely on supervised paired datasets. They lack scalability to foundation-level architectures and often focus on predictive accuracy rather than real-time multimodal reasoning or contradiction detection.

Comprehensive surveys of radiology report generation further document the evolution of these technologies. In [4], the transition from CNN–RNN encoder–decoder pipelines to transformer-based and large vision-language models is analyzed in detail. Transformer architectures enable long-range attention across image regions and textual tokens, significantly improving coherence and fluency of generated reports. Vision-Language Models and Large Language Models pretrained on massive datasets are adapted to radiology through fine-tuning. However, hallucination emerges as a critical limitation, where models generate clinically unsupported

findings. Knowledge graph-based systems attempt to address this by structuring medical relationships and guiding generation through predefined disease-anatomy associations, yet they increase architectural complexity and computational burden. The survey also highlights evaluation challenges, noting that conventional NLP metrics such as BLEU and ROUGE do not adequately capture clinical correctness, necessitating medical-specific evaluation frameworks.

Dedicated 3D report generation systems extend these developments further. In [5], a streamlined integration of a 3D vision encoder with a large language model is proposed for chest CT report generation. The architecture divides volumetric scans into 3D patches, encodes them through a transformer-based backbone, and aligns visual embeddings with a language model through a projection layer. A two-stage training strategy involving frozen encoders and low-rank adaptation is employed to address data scarcity and computational constraints. The model demonstrates substantial improvement over earlier 3D systems in text similarity metrics, confirming that large language models can be adapted effectively to volumetric imaging. Nevertheless, evaluation remains largely dependent on textual overlap metrics, and concerns regarding hallucination, modality generalization, and resource requirements persist. Similarly, [6] introduces a fully dedicated 3D sequence-to-sequence architecture for radiology report generation from chest CT volumes. This framework incorporates spatial and causal transformers to preserve volumetric continuity across slices and employs relational memory mechanisms within the decoder to maintain structured narrative consistency. An extended longitudinal variant integrates prior scans and reports to model disease progression over time. Although the model achieves improvements in both natural language generation metrics and clinical efficacy measures, it requires extensive computational resources and remains limited to specific anatomical regions.

A comprehensive survey of multimodal radiology report generation in [7] consolidates these developments into a unified pipeline encompassing multimodal input acquisition, feature learning, fusion strategies, and report generation. The survey emphasizes that most research remains centered on chest X-rays, with limited datasets for CT and MRI modalities. It identifies challenges including weak generalization across institutions, limited use of laboratory or structured clinical data, insufficient medical-correctness evaluation, and lack of interactive clinical integration. The survey concludes that automatic report generation has evolved into a

complex multimodal reasoning problem requiring coordinated advances in vision modeling, language modeling, knowledge integration, and evaluation methodology.

Collectively, prior technologies demonstrate significant progress in multimodal medical AI, particularly in automated radiology report generation and volumetric reasoning. However, most systems focus primarily on image-to-text conversion or structured captioning tasks. They rarely integrate laboratory biomarkers, structured patient metadata, and transplant allocation parameters within a unified clinical reasoning framework. Additionally, issues such as hallucination, dataset imbalance, computational expense, and limited cross-domain evaluation remain unresolved. These limitations motivate the development of an integrated system such as ScanWise AI, which extends beyond automated report drafting to incorporate multimodal correlation, contradiction detection, structured risk scoring, and transplant-aware clinical decision support within a scalable architecture.

1.4 Proposed Approach

The proposed approach of ScanWise AI is centered on the design and implementation of a multimodal clinical decision support framework that integrates medical imaging, laboratory reports, structured clinical metadata, and transplant allocation logic into a unified computational architecture. The primary aim of the project is to move beyond isolated image classification or standalone report generation systems and instead develop a correlated reasoning engine capable of synthesizing heterogeneous clinical inputs into structured, explainable outputs. The motivation for this approach arises from the fragmentation of hospital information systems, where imaging data, laboratory values, and clinical records are processed independently, requiring manual cross-verification by clinicians. By automating this cross-modal reasoning process, the system seeks to reduce diagnostic inconsistencies, improve triage prioritization, and support high-risk decisions such as organ transplantation.

The architecture follows a layered design consisting of a presentation layer, backend application layer, artificial intelligence processing layer, and correlation engine. The presentation layer is developed using a web-based interface that allows secure upload of medical images and laboratory reports. The backend server handles request routing, data validation, and structured storage while ensuring modular separation between user interaction and computational logic. The AI processing layer is responsible for extracting meaningful features from both image and text modalities. For medical imaging, a standardized

preprocessing pipeline is applied to ensure uniform input quality across different scanning machines. This includes resizing images to fixed dimensions, applying noise reduction filters, performing contrast enhancement, and normalizing pixel intensity values. These steps reduce variability introduced by hardware differences and improve model generalization.

Following preprocessing, medical images are analyzed using a Convolutional Neural Network based on transfer learning principles. A pretrained backbone model such as ResNet is fine-tuned on relevant medical imaging datasets to extract high-level spatial features representing pathological patterns. The CNN outputs probability distributions over disease categories along with confidence scores. Rather than generating free-text reports directly, the model produces structured diagnostic indicators that can be further processed by the multimodal engine. This design choice prioritizes interpretability and structured reasoning over purely generative outputs.

Parallel to image analysis, laboratory reports are processed using a Natural Language Processing pipeline. Unstructured textual data undergoes cleaning, tokenization, and Named Entity Recognition to extract biomarkers such as C-Reactive Protein, White Blood Cell count, hemoglobin levels, and other clinically relevant indicators. Extracted entities are mapped into structured key-value pairs and compared against standard clinical reference ranges. This structured representation enables numerical reasoning and threshold-based evaluation within the system.

The core component of the proposed approach is the multimodal correlation engine. This engine integrates outputs from the CNN-based imaging model and the NLP-based biomarker extraction module. A weighted reasoning mechanism is implemented to compute composite risk scores based on imaging severity, laboratory abnormalities, and patient metadata such as age and comorbidities. Logical rules are incorporated to detect contradictions, for example when imaging suggests infection but inflammatory biomarkers remain within normal limits. In such cases, the system flags the inconsistency and generates an alert for clinical review. This approach introduces cross-modal validation rather than isolated predictions.

In addition to diagnostic correlation, the proposed system incorporates a transplant matching module. This module evaluates urgency scores such as MELD and LAS, verifies ABO blood group compatibility, assesses Human Leukocyte Antigen matching, and estimates Cold Ischemia Time based on logistical parameters. A ranking algorithm combines urgency,

compatibility, and time constraints into a composite allocation score. This enables structured prioritization of transplant recipients while raising alerts if preservation time exceeds safe thresholds.

Applications of the proposed system include infection risk assessment, automated triage support, structured patient summary generation, contradiction detection between modalities, and transplant allocation prioritization. The architecture is designed to produce explainable outputs in structured formats compatible with healthcare data standards such as FHIR, enabling interoperability with Electronic Health Record systems.

1.5 Objectives

1.5.1 Behavioural Objective

The behavioural objective of the proposed system is to design and implement a multimodal clinical decision support platform capable of consistently processing heterogeneous medical inputs and producing structured diagnostic outputs within a defined operational time frame. The system must accept medical imaging data such as CT, MRI, and X-ray scans alongside laboratory reports in textual format and generate correlated clinical interpretations without requiring manual cross-verification between modalities. The behaviour of the system should demonstrate deterministic and repeatable outputs for identical inputs, ensuring reliability in clinical environments. It must compute disease probability scores from imaging data, extract biomarker values from laboratory reports, and produce composite risk assessments within an average inference time of approximately five seconds per case under standard hardware conditions.

The behavioural design also requires the system to detect logical inconsistencies between modalities, such as mismatches between radiological findings and inflammatory markers, and automatically generate alerts for clinician review. These behavioural characteristics must be demonstrable through controlled testing scenarios where predefined cases produce predictable risk scores and structured summaries.

The behavioural objective also requires the system to maintain stable and predictable outputs under repeated testing conditions. For identical inputs, the generated diagnostic scores and summaries must remain consistent to ensure clinical reliability. The system should

appropriately scale risk levels when variations occur in imaging findings or biomarker values. Additionally, it must clearly differentiate between high-confidence and uncertain predictions, reflecting uncertainty in a structured and transparent manner suitable for clinical review.

1.5.2 Analytical Objective

The analysis objective focuses on implementing and validating a multimodal correlation engine capable of performing structured reasoning across imaging features, laboratory biomarkers, and patient metadata. The system must extract high-level visual features using a Convolutional Neural Network and convert unstructured clinical text into structured numerical variables through Natural Language Processing techniques. These outputs are then integrated using a weighted scoring mechanism that computes composite diagnostic risk levels. The analytical capability of the system must be measurable through internal validation tests that assess agreement between imaging-derived severity indicators and laboratory abnormality thresholds.

The objective includes demonstrating the system's ability to quantify disease likelihood, detect contradictions between modalities, and produce interpretable risk scores rather than free-form generative outputs. Furthermore, the analytical module must incorporate transplant-related parameters such as urgency scoring and compatibility checks, enabling objective prioritization logic. The analytical performance should be demonstrable through simulation of multiple clinical scenarios, confirming that the correlation engine produces consistent and clinically coherent outputs across diverse input combinations.

1.5.3. System Management Objective

The system management objective is to develop a modular and maintainable software architecture that ensures structured data flow, separation of concerns, and scalability. The frontend interface, backend server, artificial intelligence modules, and correlation engine must function as independent yet interoperable components. The objective includes implementing well-defined APIs for communication between layers, structured data schemas for image and laboratory inputs, and asynchronous processing mechanisms to prevent bottlenecks during inference. The system must demonstrate organized code structure, including separation of model logic, utility functions, and service modules. Error handling routines must be incorporated to manage invalid inputs, corrupted image files, or incomplete laboratory reports without system failure. This objective ensures that the project is not only algorithmically

functional but also engineered for maintainability and future extension. The system should be capable of supporting additional modalities or updated algorithms without requiring fundamental redesign of the architecture.

1.5.4 Security Objective

The security objective is to ensure that all patient-related data processed by the system is handled with appropriate safeguards and structured validation mechanisms. The system must implement secure file upload handling, input sanitization, and controlled access to backend services. Structured output formatting should comply with standardized healthcare interoperability frameworks to facilitate integration while maintaining data integrity. The objective includes implementing validation checks to prevent injection of malicious data through textual laboratory inputs and ensuring that image files are processed in a controlled environment. Sensitive clinical variables must not be exposed in unsecured logs or responses. This objective is demonstrable through system-level testing that verifies safe handling of incorrect file types, malformed text, or unauthorized access attempts. By embedding security controls at the architectural level, the system aligns with healthcare data protection requirements and ensures safe operational deployment.

1.5.5 Deployment Objective

The deployment objective is to develop a prototype that can be executed in a cloud-enabled or local server environment with interoperability readiness for integration into hospital information systems. The system must demonstrate the ability to run inference on uploaded medical scans and laboratory reports through a deployed interface accessible via a secure web endpoint. Structured outputs should be generated in standardized formats compatible with Electronic Health Record systems to support potential real-world integration. The deployment pipeline must include model loading, API configuration, and performance monitoring to ensure stable runtime behavior. The objective also requires documenting hardware requirements, inference time benchmarks, and resource utilization metrics. Demonstrable deployment includes successfully hosting the application in a test environment where multiple user requests can be processed without performance degradation. This ensures that the project extends beyond theoretical modeling and represents a practically deployable multimodal clinical intelligence system.

1.6 Sustainable Development Goals (SDGs)

This project contributes directly to several United Nations Sustainable Development Goals (SDGs):

- Goal 3: Good Health and Well-Being

The project supports Goal 3 by enhancing the quality, speed, and reliability of medical diagnosis through an AI-powered multimodal decision support system. By integrating medical imaging, laboratory reports, and clinical metadata into a unified analytical framework, the system assists clinicians in early detection of diseases, identification of diagnostic inconsistencies, and prioritization of high-risk cases. Faster triage and structured risk scoring can reduce delays in treatment, particularly in critical conditions such as infections and organ failure. By supporting safer transplant allocation decisions and minimizing diagnostic errors, the system contributes to improved patient outcomes and overall healthcare effectiveness.

- Goal 9: Industry, Innovation, and Infrastructure

The project supports Goal 9 by introducing advanced artificial intelligence technologies into healthcare infrastructure. It combines deep learning, natural language processing, and structured clinical reasoning within a scalable and interoperable system architecture. By aligning with digital health standards and enabling integration with electronic health record systems, the project strengthens healthcare infrastructure through innovation. The use of modular design and cloud-deployable architecture promotes technological modernization in hospitals, contributing to resilient, efficient, and data-driven healthcare systems.

- Goal 10: Reduced Inequalities

The project supports Goal 10 by addressing disparities in access to specialized medical expertise. In many regions, particularly rural or resource-limited areas, there is a shortage of radiologists and transplant specialists. An AI-assisted diagnostic system can help bridge this gap by providing structured preliminary analysis and decision support. By standardizing interpretation across imaging and laboratory data, the system reduces

dependency on limited specialist availability and promotes more equitable healthcare delivery. This can improve access to quality diagnostic support regardless of geographic location.

- Goal 17: Partnerships for the Goals

The project supports Goal 17 by encouraging collaboration between healthcare institutions, technology developers, and research communities. The system is designed for interoperability with hospital information systems and structured healthcare data standards, enabling integration across departments and institutions. Successful deployment of such a platform requires coordination among clinicians, engineers, policymakers, and data governance authorities. By fostering multidisciplinary cooperation and technological integration, the project aligns with the goal of building strong partnerships to achieve sustainable development objectives in healthcare.



Fig 1.1 Sustainable Development Goals

1.7 Overview of Project Report

Chapter 1 introduces the project, explaining its motivation, background, objectives, and alignment with the SDGs. Chapter 2 presents the literature review, covering existing work on microgrid cybersecurity, encryption methods, and machine learning-based intrusion detection. Chapter 3 describes the methodology and system architecture, while Chapter 4 discusses the

implementation details and software configuration. Chapter 5 focuses on analysis and design, including requirements and flow diagrams. Chapter 6 outlines the software development tools. Chapter 7 explains the testing methodology, test plans, and results obtained. Chapter 8 discusses social, legal, ethical, sustainability, and safety aspects related to the project. Finally, Chapter 9 presents the conclusion, summarizing findings and suggesting future improvements.

Chapter 2

LITERATURE REVIEW

Hosseini, Ibrahim, and Serag propose the M3 framework, a multimodal artificial intelligence system designed for medical report generation and visual question answering from 3D abdominal CT scans. The study addresses a key limitation in earlier vision-language models that primarily focused on 2D images such as chest X-rays, ignoring the spatial continuity present in volumetric data. Built on the AMOS-MM dataset, the framework incorporates dataset refinement strategies to handle imbalance and short-caption bias. The architecture consists of a 3D vision encoder (M3D), a projection layer for cross-modal alignment, and a language model initialized from Phi-3. Training involves fine-tuning with controlled learning rates and temperature settings to improve clinical consistency. Evaluation uses the GREEN score to measure factual correctness rather than simple text overlap. Although the framework demonstrates that 3D multimodal reasoning is feasible, performance remains moderate, and challenges such as dataset imbalance, evaluation bias, and high computational cost persist.

Lee et al. present a conceptual framework for extending multimodal generative AI from 2D images to 3D medical scans and long procedural videos. The authors propose treating 3D CT and MRI volumes as sequential frames similar to videos, enabling adaptation of video-text transformer architectures. A major contribution is the concept of self-multimodality, where multiple imaging phases or sequences are interpreted jointly rather than independently. The paper emphasizes the importance of metadata, scan protocols, and clinical context as integral inputs to AI systems. Training strategies such as contrastive alignment and masked autoencoding are discussed to enhance cross-modal learning. Applications include automated report drafting and real-time procedural guidance. However, the work highlights critical limitations including data scarcity, privacy concerns, lack of standardized benchmarks, and computational demands. The study serves as a roadmap for future 3D multimodal systems rather than presenting a finalized deployable model.

Yao et al. introduce a multimodal deep learning framework that integrates medical images and clinical reports for improved disease diagnosis and lesion localization. The architecture combines a CNN-based image encoder with a Bi-LSTM text encoder enhanced by attention mechanisms. Visual and textual features are fused through a multimodal interaction layer to

create a shared representation for downstream tasks. The model is evaluated on disease classification, lesion localization, and clinical description generation, outperforming single-modality baselines. Results show improved accuracy, recall, and F1-score when both modalities are jointly analyzed. The study demonstrates that deep feature fusion enhances diagnostic reliability by aligning spatial evidence with semantic interpretation. However, the approach relies on supervised paired datasets and task-specific architectures, limiting scalability compared to newer foundation-model approaches.

Mamdouh et al. provide a comprehensive survey of radiology report generation techniques between 2021 and 2025, documenting the evolution from CNN–RNN pipelines to transformer-based and vision-language models. The survey categorizes approaches into text summarization and image-to-text generation and discusses advancements in encoder–decoder architectures. It highlights the increasing role of large language models and knowledge graph–based systems in improving report fluency and structure. The authors identify hallucination as a major challenge, where models generate unsupported findings. Evaluation limitations are also discussed, noting that traditional NLP metrics do not adequately capture clinical correctness. The survey emphasizes the need for improved benchmarks, generalization across datasets, and interpretability before clinical deployment. This work provides a structured understanding of the current state and open challenges in automated radiology reporting.

Chen et al. propose 3D-CT-GPT, a vision-language model designed for automated report generation from 3D chest CT volumes. The system integrates a 3D vision encoder (CT-ViT) with a large language model through a projection layer that aligns visual and textual embeddings. A two-stage training strategy involving frozen encoders and low-rank adaptation enables efficient fine-tuning under limited data conditions. The model is evaluated on CT-RATE and private datasets using standard language generation metrics, demonstrating significant improvements over earlier 3D systems. The approach shows that volumetric data can be effectively processed using transformer-based architectures combined with large language models. Despite strong results, the model remains computationally intensive and dependent on text similarity metrics for evaluation. The study establishes an efficient pathway for adapting LLMs to 3D medical imaging.

Hamamci et al. introduce CT2Rep, the first end-to-end framework dedicated to radiology report generation from 3D chest CT volumes. The model uses a patch-based 3D transformer

encoder combined with a memory-enhanced transformer decoder. A two-stage spatial and causal transformer design preserves volumetric continuity across slices, while relational memory improves narrative consistency in generated reports. An extended version, CT2RepLong, incorporates longitudinal data to model disease progression over time. Evaluation includes both natural language generation metrics and clinical efficacy scores. Results demonstrate improved report coherence and abnormality detection compared to classification-based baselines. However, training requires substantial computational resources, and inference time remains high. The work establishes a foundational architecture for volumetric report generation.

Wang et al. present a detailed survey of multimodal deep learning approaches for radiology report generation, reviewing 89 studies published between 2021 and 2024. The authors propose a unified five-stage pipeline covering input acquisition, preprocessing, feature learning, fusion, and report generation. The survey highlights the transition from CNN-based encoders to hybrid CNN-Transformer and pure Transformer architectures. It also discusses contrastive learning, memory mechanisms, and knowledge graph integration for improved multimodal alignment. Evaluation challenges are emphasized, particularly the gap between text similarity metrics and true medical correctness. The survey identifies limited use of non-image clinical data and insufficient benchmark diversity as key research gaps. This work frames automated reporting as a multimodal reasoning problem rather than simple image captioning.

Durairaj and Uthirapathy propose the Trans-IMSM model for CT-MRI image fusion aimed at improving brain tumor detection. The framework combines guided-filter-based image decomposition with transformer-driven multi-scale and multi-modal attention mechanisms. By separating base and detail layers, the model preserves both structural integrity and fine-grained tumor boundaries. Cross-modal attention allows CT and MRI features to interact dynamically rather than being fused through simple concatenation. A gated reconstruction network adaptively balances fused components. Experimental results demonstrate high accuracy and strong structural similarity compared to baseline fusion methods. The study highlights the importance of multi-scale and cross-modal interaction in medical image fusion. However, computational complexity and dataset size limit immediate clinical deployment.

Aksoy et al. propose an integrative multimodal framework for chest X-ray report generation that incorporates imaging data, structured clinical variables, demographic metadata, and

unstructured text inputs. Unlike image-only systems, their model uses cross-attention mechanisms to dynamically align patient context with visual features. The dataset is constructed by linking multiple clinical databases to ensure synchronized imaging and non-imaging data. An EfficientNet-based image encoder and transformer decoder are used for generation. Evaluation using BLEU, ROUGE-L, and clinical BERT-based metrics shows improved performance over image-only baselines. Human radiologist evaluation confirms improvements in content relevance and fluency, although hallucination and subtle diagnostic errors remain. The study demonstrates that integrating structured clinical data significantly enhances the realism and quality of generated reports.

Table 2.1 Summary of Literatures Reviewed

S#	Article Title, Published year, Journal name	Methods	Key features	Merits	Demerits
1	M3: Multimodal AI for Medical Report Generation and VQA from 3D Abdominal CT Scans, 2025, BJR AI	3D encoder (M3D), CLIP-style contrastive learning, Phi-3 LLM fine-tuning	Handles 3D CT, report generation + VQA, spatial pooling of slices	First strong 3D multimodal system, supports reasoning tasks	Dataset imbalance, high compute, evaluation metric (GREEN) weak
2	Multimodal Generative AI for Interpreting 3D Medical Images and Videos, 2025, npj Digital Medicine	Video-text model adaptation, contrastive learning, masked autoencoding	Treats 3D scans as videos, supports multiphase data, metadata integration	Strong conceptual roadmap, supports long sequences	No complete system, lacks large datasets, heavy compute
3	Integrating Medical Imaging and Clinical Reports Using Multimodal DL, 2024	CNN for images, Bi-LSTM with attention for text, fusion layer	Joint image-text learning, disease classification + localization + report gen	High accuracy (96.42%), interpretable modular design	Limited dataset details, heavy computation
4	Advancements in Radiology Report Generation: A Comprehensive Analysis, 2025	Survey of CNN-RNN, Transformers, VLMs, LLMs, graph-based models	Covers datasets, metrics, hallucination control	Best overview of 2021–2025 RRG	No new model, limited empirical results
5	3D-CT-GPT: Generating 3D Radiology Reports with VLMs, 2024	CT-ViT encoder, projection layer, Vicuna-7B, LoRA fine-tuning	Efficient 3D patch embedding, staged training	State-of-art 3D report gen, efficient training	Hallucination risk, chest-only data

6	CT2Rep: Automated Radiology Report Generation for 3D Imaging, 2024	3D patch transformer, causal transformer, relational memory	First end-to-end 3D report generator, longitudinal fusion	Strong clinical accuracy, supports prior scans	Very slow, heavy GPU use
7	Survey of Deep Learning-Based RRG Using Multimodal Data, 2024, arXiv	Review of CNN, Transformers, contrastive learning, memory models	Unified 5-stage pipeline, multimodal focus	Best technical organization of field	No experiments, survey only
8	Interactive Multi-scale Fusion for Brain Tumor Detection (Trans-IMSM), 2025	Guided filtering, multi-scale transformer, cross-attention	Separates base/detail layers, CT-MRI interaction	98.64% accuracy, strong fusion quality	Heavy model, low-res dataset
9	Beyond Images: Multi-Modal Chest X-ray Report Generation, 2024, Frontiers in Radiology	EfficientNet + Transformer, cross-attention fusion	Uses vitals, text, demographics with images	Best ROUGE-L, radiologist validated	Single-institution data, heavy compute
10	Multimodal Generative AI for Interpreting 3D Medical Images and Videos, 2025, <i>npj Digital Medicine</i>	Video-text model adaptation, contrastive learning, masked autoencoding	Treats 3D scans as videos, supports multiphase data	Strong conceptual framework, handles long sequences	No complete system, high compute demand

Chapter 3

METHODOLOGY

3.1 Overview of the Development Methodology

The development of ScanWise AI - AI-Powered Multi-Scan Medical Imaging and Report Interpretation System follows a structured V-Model Software Development Methodology, emphasizing systematic design, verification, and validation at each stage of development. The V-Model is particularly suitable for healthcare-oriented artificial intelligence systems because it ensures that every design decision is traceable to defined clinical and technical requirements. Given that ScanWise integrates medical imaging analysis, laboratory report processing, multimodal correlation, and transplant allocation logic, a disciplined development lifecycle was necessary to ensure reliability, interpretability, and safety.

The descending side of the V represents progressive decomposition of the system into smaller and manageable components, beginning from high-level clinical requirements down to individual algorithmic units such as CNN modules and NLP pipelines. The ascending side of the V corresponds to integration, verification, and validation processes that confirm whether each implemented component satisfies predefined requirements. This structured approach is essential in healthcare systems, where incorrect outputs may directly influence clinical decisions.

Unlike generic software systems, ScanWise combines data-driven learning modules with rule-based clinical reasoning. This hybrid architecture requires careful testing at both the algorithmic and system levels. The V-Model ensures that model training, feature extraction, multimodal correlation, and transplant ranking logic are not treated as isolated modules but as interconnected components that must collectively satisfy clinical objectives.

The methodology also incorporates iterative refinement within each stage. While the V-Model provides a structured framework, internal iterations were performed during model optimization, parameter tuning, and rule calibration. This ensured that performance improvements did not compromise interpretability or stability. The result is a traceable,

verifiable, and clinically aligned development workflow suitable for AI-based medical decision support systems.

3.2 Mapping of Project Phases to the V-Model

The V-Model for ScanWise consists of the following phases: requirement analysis, system design, functional design, unit design and implementation, unit testing, integration testing, verification, and validation. Each phase corresponds to a distinct stage in building the multimodal clinical decision support framework.

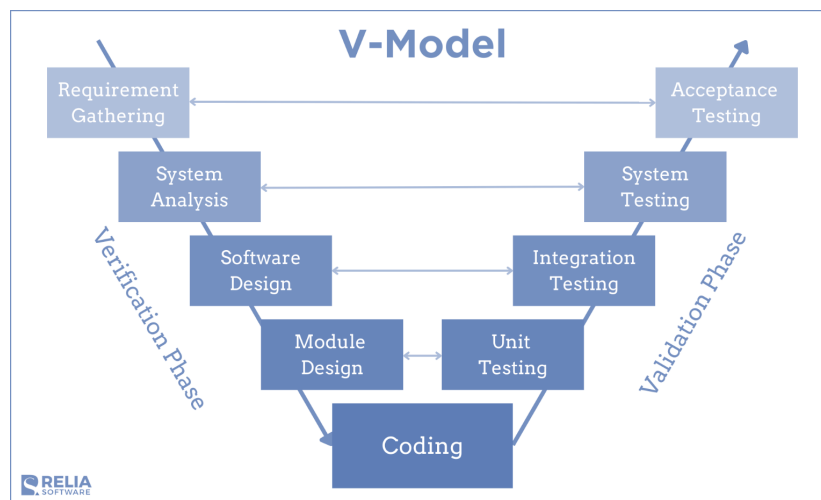


Fig 3.1 V-Model phase mapping

3.2.1 Requirement Analysis

The requirement analysis phase focused on identifying functional, clinical, and system-level requirements for a multimodal AI platform capable of integrating imaging data, laboratory reports, and transplant allocation logic. Literature on radiology report generation, multimodal AI systems, and clinical decision support tools informed the requirement definition process. Functional requirements included automated preprocessing of CT, MRI, and X-ray images; structured extraction of laboratory biomarkers from unstructured reports; computation of disease probability scores; detection of cross-modal inconsistencies; generation of structured patient summaries; and transplant prioritization based on urgency and compatibility metrics. Non-functional requirements included inference time below five seconds per case, structured output compatibility with FHIR standards, modular scalability, and secure handling of sensitive patient data.

The analysis phase also identified technical constraints. Medical images may vary in resolution, orientation, and quality depending on acquisition devices. Laboratory reports are often unstructured and inconsistent in formatting. Therefore, preprocessing normalization and text standardization mechanisms were defined as mandatory requirements. Additionally, the system needed to ensure deterministic outputs for identical inputs to maintain clinical reliability.

Performance metrics were defined during this phase to measure system effectiveness. For imaging models, classification accuracy, precision, recall, and confidence calibration were selected. For NLP extraction, entity recognition accuracy and numerical value extraction consistency were defined. For the correlation engine, contradiction detection accuracy and composite risk score stability were established as evaluation criteria. This phase established a comprehensive requirement baseline that guided all subsequent design and implementation stages.

3.2.2 System Design

The system design phase defined the architectural framework of the ScanWise platform by organizing the system into distinct but interrelated layers. The architecture consists of four major layers: the user interface layer, the backend orchestration layer, the AI processing layer, and the multimodal reasoning layer. The user interface layer enables clinicians to upload medical scans and laboratory reports securely while visualizing structured outputs such as disease probability scores, extracted biomarker values, and transplant priority levels. The backend orchestration layer is responsible for managing request handling, validating input formats, routing data to appropriate processing modules, and ensuring consistent response formatting. The AI processing layer includes two independent subsystems, one dedicated to imaging analysis and the other to natural language processing. The multimodal reasoning layer integrates outputs from these subsystems using predefined scoring logic and validation rules. This layered separation ensures controlled data flow and reduces interdependency between modules. Communication between layers is governed by standardized data schemas to prevent inconsistencies. Each layer performs a well-defined computational role within the system.

The architectural design supports asynchronous execution of independent modules. Imaging analysis and laboratory report processing can occur in parallel, improving computational efficiency. Synchronization occurs only at the reasoning stage where outputs are merged. This reduces processing latency while maintaining structural alignment of data.

Within the AI processing layer, the imaging subsystem was structured around a convolutional neural network pipeline with integrated preprocessing routines. Input images undergo resizing, normalization, and contrast adjustment before feature extraction. The CNN architecture was selected to support transfer learning, enabling future extension to additional scan categories. Output from the classifier is formatted as a structured probability vector with calibrated confidence scores.

The NLP subsystem processes laboratory reports through tokenization, text normalization, entity recognition, and numerical value extraction. Extracted biomarker values are mapped against predefined reference intervals to determine abnormality status. These outputs are converted into standardized dictionaries before being forwarded to the reasoning module. The multimodal reasoning layer applies weighted arithmetic combinations and logical consistency checks. This ensures that final interpretations are based on aligned evidence from both modalities.

The system design also incorporates structured logging mechanisms at each layer. Intermediate outputs from imaging, NLP, and reasoning modules are recorded to maintain traceability. Error-handling routines are embedded within module boundaries to manage incomplete inputs or parsing failures. Data validation checkpoints prevent corrupted or inconsistent records from propagating across layers. These mechanisms strengthen operational reliability without altering computational logic.

3.2.3 Functional Design

The functional design phase decomposed each architectural layer into clearly defined computational functions with explicit input and output contracts. The imaging module was divided into preprocessing, feature extraction, classification, and confidence adjustment stages. Preprocessing functions include resolution normalization, pixel intensity scaling, and noise suppression to standardize incoming scan data. The feature extraction stage uses convolutional layers to capture spatial patterns and abnormal regions.

The classification stage produces probability distributions across defined disease categories. A calibration step adjusts output probabilities to reduce overconfidence bias. Each function was implemented as an independent callable unit to allow isolated testing and refinement. Data

structures were standardized to ensure compatibility with downstream modules. The output from this pipeline is a structured vector representing disease likelihoods.

The NLP module was functionally divided into text cleaning, entity identification, numerical parsing, abnormality determination, and structured representation generation. Laboratory text undergoes tokenization and normalization before named entity recognition identifies biomarkers. Extracted numerical values are validated for format consistency and compared against reference ranges. Each biomarker is assigned a status label indicating normal or abnormal range.

The multimodal reasoning module combines structured outputs from the imaging and NLP modules using weighted scoring logic. Imaging probability values are scaled and combined with biomarker abnormality indicators to compute composite risk scores. Logical validation rules evaluate consistency between modalities and flag contradictory evidence. For transplant prioritization, urgency scores are computed based on severity indices and compatibility factors. Each computational step uses deterministic arithmetic operations to ensure reproducibility. Intermediate scoring variables are stored before final decision computation. The reasoning engine does not modify upstream outputs but applies structured transformations. This separation maintains transparency of original model predictions while enabling integrated interpretation.

3.2.4 Unit Design and Implementation

In the unit design and implementation phase, each functional component of the ScanWise system was developed as an independent computational module with clearly defined responsibilities. The imaging pipeline was implemented as a sequence of callable units including preprocessing, CNN inference, probability calibration, and structured output formatting. Each stage produces a well-defined intermediate output, ensuring traceability of predictions. The NLP module was implemented as a separate pipeline consisting of text cleaning, entity recognition, numerical extraction, reference comparison, and abnormality tagging. Extracted biomarker values were stored in structured dictionaries to maintain uniform data representation.

The multimodal reasoning engine was implemented as a deterministic scoring module that consumes structured outputs from both imaging and NLP subsystems. All modules were

designed to operate independently and communicate through standardized data objects. This modular design reduced coupling between subsystems and simplified maintenance.

The transplant prioritization module was implemented as a rule-based scoring engine that computes urgency and compatibility indices. Risk levels from the reasoning engine are mapped into transplant priority categories using predefined thresholds. Compatibility constraints are verified through logical checks before final ranking is generated.

During implementation, strict input-output contracts were enforced between modules to prevent inconsistent data propagation. Each unit performs validation checks on incoming data before processing, ensuring that malformed images or improperly formatted laboratory reports are rejected early. Logging mechanisms were embedded within each unit to record intermediate computations and detected anomalies. The CNN module was implemented using transfer learning techniques to improve convergence stability. The NLP module integrates pretrained medical language models for accurate entity recognition.

The reasoning engine uses fixed weighted coefficients that can be adjusted without modifying upstream AI modules. These implementation decisions preserve transparency while maintaining flexibility for refinement. Asynchronous execution capability was incorporated so imaging and NLP pipelines could operate concurrently before synchronized fusion. Data serialization formats were standardized to ensure compatibility between backend services and AI modules. Error-handling routines were embedded within each function to prevent cascade failures. Intermediate confidence scores and abnormality flags are stored before final interpretation. This structure enables precise tracing of decisions during later verification stages.

3.2.5 Unit Testing

Each module of the ScanWise system was evaluated independently to verify correctness and performance before integration. The imaging module was tested using validation datasets to measure classification accuracy, precision, recall, and calibration stability. Preprocessing routines were tested separately to ensure consistent resizing and normalization across varied image formats. The NLP module was tested using labeled laboratory reports to validate entity extraction accuracy and numerical parsing consistency. Biomarker abnormality classification was verified against predefined reference ranges to confirm threshold logic correctness. The

multimodal reasoning engine was tested with synthetic combinations of imaging probabilities and biomarker abnormalities to verify weighted scoring accuracy. Transplant prioritization calculations were validated against predefined urgency scenarios to ensure deterministic ranking behavior. These unit-level tests ensured that each computational block behaved as intended under controlled inputs.

Testing also examined input validation routines to confirm that incomplete or malformed data was correctly rejected. Edge cases such as missing biomarker values, low-confidence imaging outputs, or conflicting modality results were injected to evaluate system robustness.

Performance evaluation was conducted in isolation to measure computational efficiency and stability of each module. Imaging inference time was measured to ensure responsiveness within acceptable limits. NLP processing time was evaluated to confirm efficient parsing of lengthy reports. The reasoning engine was stress-tested with repeated data combinations to verify numerical stability and absence of overflow or scaling issues. Calibration tests were performed to ensure probability outputs remained within expected ranges. Memory usage and computational load were monitored to detect potential inefficiencies.

3.2.6 Integration Testing

In the integration testing phase, all independently validated modules of the ScanWise system were combined to evaluate coordinated system behavior. The imaging pipeline, NLP extraction module, multimodal reasoning engine, transplant prioritization module, and backend orchestration services were integrated into a unified workflow. Data synchronization between modules was tested to ensure consistent formatting and correct sequencing of operations. Imaging and laboratory inputs were processed simultaneously to confirm that asynchronous execution did not introduce inconsistencies at the fusion stage.

Structured outputs from each subsystem were monitored to verify compatibility before multimodal scoring. Logging systems were evaluated to ensure that intermediate outputs remained traceable across module boundaries. Integration testing also confirmed that the backend correctly handled parallel requests without data collision. Controlled datasets were passed through the full pipeline to evaluate whether combined outputs aligned with expected results. This stage ensured that independent modules operated cohesively as a single decision-support system.

Test scenarios included normal multimodal inputs, incomplete laboratory reports, low-confidence imaging predictions, and intentionally mismatched modality combinations. These scenarios assessed whether the reasoning engine correctly flagged inconsistencies and handled missing data without computational failure. Transplant prioritization logic was tested under varying severity conditions to confirm deterministic ranking behavior.

Beyond standard integration, stress scenarios were introduced to examine how the system handled compounded irregularities. For example, low imaging confidence combined with abnormal biomarkers was used to evaluate weighted scoring adjustments. Simultaneous high-risk indicators across modalities were injected to verify escalation behavior within the transplant module. Deliberate formatting inconsistencies were introduced to confirm that input validation routines prevented propagation of malformed data.

The system's response time was monitored under repeated batch processing to assess performance stability. Logging consistency across modules was evaluated to confirm synchronized recording of intermediate outputs. These tests ensured that integration did not introduce latency spikes, misalignment of outputs, or unintended computational dependencies.

Data flow consistency was verified by tracking structured outputs at each processing stage. Error-handling responses were evaluated to confirm graceful failure when critical inputs were absent. Cross-module communication logs were reviewed to detect unexpected sequencing behaviour. These integration checks ensured reliable coordinated execution across all system layers.

3.2.7 Verification

Verification activities were conducted to confirm that each implemented feature met its originally defined functional and performance requirements. The imaging subsystem was verified by comparing predicted disease classifications against labeled validation datasets to ensure alignment with accuracy targets. Probability calibration was verified by examining output distribution stability across repeated inference cycles. The NLP subsystem was verified by measuring entity extraction accuracy and validating numerical parsing consistency across structured reports. Biomarker abnormality detection was confirmed through comparison against established medical reference ranges.

The multimodal reasoning engine was verified by manually computing weighted scoring examples and confirming that automated outputs matched expected results. Transplant prioritization logic was verified by testing predefined urgency cases to confirm deterministic ranking behavior. These verification steps ensured that each computational output adhered to predefined specifications.

Performance thresholds defined during requirement analysis were systematically evaluated. Imaging inference time was measured against predefined latency constraints. NLP extraction performance was verified to maintain structured output consistency across diverse report formats. Logical consistency checks within the reasoning engine were verified to correctly identify contradictory modality signals.

3.2.8 Validation

The validation stage evaluated the fully integrated ScanWise system against the predefined clinical and technical objectives of the project. End-to-end multimodal workflows were executed using combined medical imaging scans and laboratory reports to assess real-time interpretative performance. The system's ability to generate structured disease probability scores, accurately extract biomarkers, and compute transplant priority rankings was examined under varied input conditions. Cross-modal inconsistency detection was tested using deliberately conflicting imaging and laboratory data.

Response time and computational stability were monitored to ensure practical usability. Repeated identical inputs were processed to verify deterministic behavior and output consistency. The results demonstrated stable multimodal integration, reliable scoring behavior, and transparent traceability of intermediate computations. These outcomes confirmed that the system operates consistently within its defined decision-support scope and aligns with the original development objectives.

Additional validation involved evaluating output interpretability to ensure that probability scores, abnormality flags, and risk levels were clearly understandable. Edge cases involving incomplete laboratory data were tested to confirm graceful handling without computational failure.

3.3 Overview of the Proposed System Implementation

ScanWise AI is a multimodal clinical decision-support system that integrates medical imaging, laboratory reports, and patient metadata within a unified architecture. The system processes uploaded scans through preprocessing and CNN-based analysis, while laboratory reports are interpreted using structured NLP extraction. Outputs from both modalities are forwarded to a correlation engine that computes risk scores and detects inconsistencies. The architecture follows a layered pipeline including ingestion, AI inference, reasoning, and presentation. This structured implementation ensures modularity, interpretability, and reliable clinical support.

3.3.1 Architecture Overview

The ScanWise AI architecture is composed of seven major components: Scan Ingestion Module, Image Pre-Processing Module, AI Diagnosis Module, Report Interpretation Module, Agentic Decision Support Module, Clinician Dashboard, and Patient Communication Module. These modules operate sequentially yet maintain bidirectional data consistency through structured JSON-based internal representations.

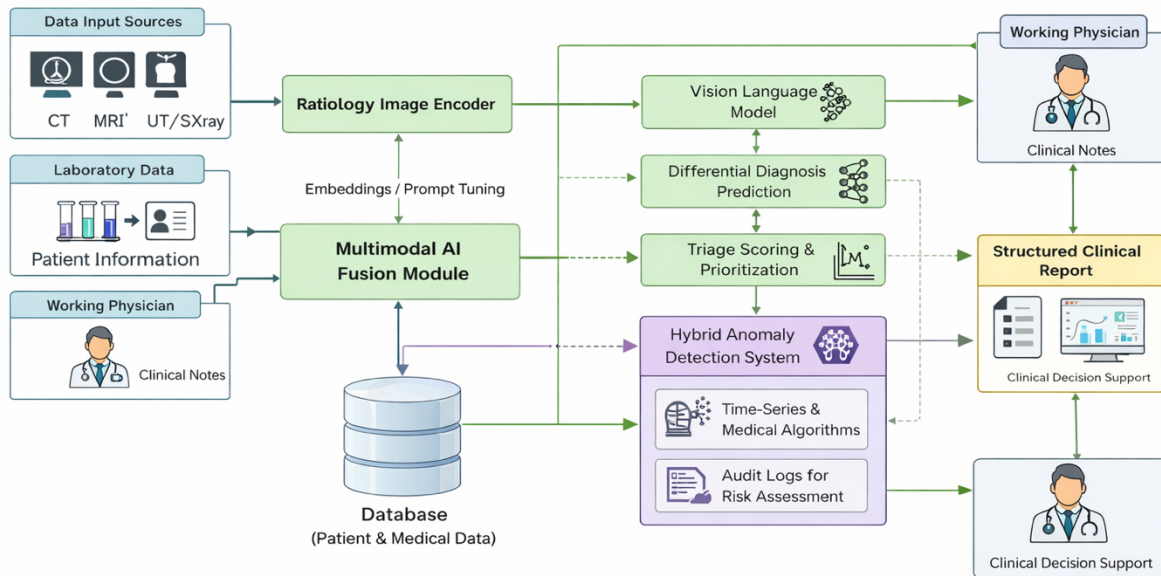


Fig 3.2 System architecture

The system begins with secure ingestion of DICOM or image-based radiological scans and laboratory reports. The ingestion layer standardizes formats and validates metadata before forwarding them to pre-processing pipelines. Imaging data flows into convolutional neural network inference engines, while textual data is processed through structured NLP extraction

pipelines. Outputs from both modalities are forwarded to the multimodal reasoning engine, where weighted risk scoring, contradiction detection, and triage prioritization are computed. Final outputs are then rendered through separate clinician and patient interfaces with different abstraction levels.

The AI processing core functions as the computational backbone of the system. It maintains separation between feature extraction and reasoning layers. Feature extraction includes CNN-based spatial encoding and NLP-based semantic extraction. The reasoning layer implements weighted scoring and rule-based logic for clinical validation. This separation ensures interpretability and controlled decision-making rather than black-box output generation.

3.3.2 AI Diagnosis Module (CNN-Based Medical Image Analysis)

The AI Diagnosis Module forms the primary computational engine responsible for automated interpretation of radiological scans including CT, MRI, and X-ray images. This module implements a deep convolutional neural network architecture based on transfer learning principles, where a pretrained Residual Network (ResNet) backbone is fine-tuned for medical imaging tasks. Transfer learning is adopted to leverage pretrained feature representations while adapting higher convolutional layers to domain-specific radiological patterns. This approach reduces training time, improves convergence stability, and mitigates overfitting under limited medical dataset conditions.

The convolutional operation used within the network is defined as:

$$S(i, j) = \sum_m \sum_n I(i - m, j - n) K(m, n)$$

where I represents the input image matrix, K denotes the convolution kernel, and $S(i, j)$ is the resulting feature map. Through stacked convolutional layers, the network extracts hierarchical features beginning from low-level edges and gradients to higher-order anatomical patterns such as nodules, consolidations, or structural asymmetries.

Residual connections are incorporated to address vanishing gradient problems in deep architectures. The residual mapping is expressed as:

$$y = F(x, W) + x$$

where $F(x, W)$ represents the learned residual function and x is the identity mapping. This formulation allows stable gradient propagation during backpropagation and enables deeper networks without degradation in accuracy.

The final classification layer applies the Softmax activation function to generate probabilistic outputs across diagnostic classes:

$$aP(y = i | z) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

where z_i is the logit for class i and C represents the total number of diagnostic categories. The model outputs both the predicted class and a calibrated confidence score, which is later used by the multimodal reasoning engine to compute weighted risk assessments.

Loss optimization during training is performed using categorical cross-entropy:

$$\mathcal{L} = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

where y_i is the ground truth label and y^i is the predicted probability. This formulation ensures probabilistic alignment between predictions and labeled outcomes.

To enhance interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is implemented to visualize regions of interest contributing to classification decisions. This enables clinicians to verify whether the network focuses on clinically meaningful anatomical structures rather than irrelevant image regions, strengthening trust within the diagnostic workflow.

3.3.3 Image Pre-Processing Module

The Image Pre-Processing Module is responsible for transforming heterogeneous radiological inputs into standardized tensors suitable for deep learning inference. Medical scans acquired from CT, MRI, and X-ray systems differ in resolution, intensity distribution, acquisition protocol, and noise characteristics. Without normalization, these inconsistencies significantly degrade model generalization. Therefore, preprocessing is not merely a formatting step but a statistical conditioning stage that ensures stable feature extraction.

The first operation is spatial standardization. Each image is resized to a fixed dimension, typically $224 \times 224 \times 224$, to match the input requirement of the CNN backbone. Resizing is implemented using bilinear interpolation, preserving structural continuity while avoiding aliasing artifacts. Following resizing, intensity normalization is applied to reduce variance across acquisition devices. Pixel scaling is performed as:

$$x_{\text{norm}} = \frac{x - \mu}{\sigma}$$

where μ and σ represent the mean and standard deviation of pixel intensities. This standardization centers the distribution and improves gradient stability during inference.

To suppress high-frequency acquisition noise, Gaussian smoothing is applied using a convolution kernel defined by:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

This reduces sensor noise while preserving anatomical boundaries. Contrast enhancement via histogram equalization is applied selectively to enhance low-contrast regions such as soft tissue boundaries. The preprocessing pipeline ensures that subsequent CNN layers operate on statistically consistent and clinically meaningful representations.

3.3.4 Report Interpretation Module (Natural Language Processing Engine)

The Report Interpretation Module is responsible for converting unstructured or semi-structured clinical text into structured, computable representations. Laboratory reports and radiology narratives often contain biomarker values, reference ranges, interpretative notes, and contextual observations embedded within free-form text. Manual cross-referencing of these values is time-consuming and prone to oversight. This module automates semantic extraction while preserving clinical context.

The pipeline begins with textual normalization, including removal of formatting artifacts, standardization of measurement units, and resolution of abbreviation inconsistencies. Medical terminology is inherently domain-specific, and the module incorporates vocabulary normalization strategies to map synonymous expressions into canonical forms. For example,

“C-Reactive Protein” and “CRP” are mapped to the same internal biomarker representation. Unit harmonization ensures that values such as mg/dL and mg/L are converted into consistent scales before interpretation.

Tokenization and syntactic parsing are then applied to segment sentences and identify candidate entities. Named Entity Recognition (NER) models are employed to detect biomedical entities including biomarkers, numerical values, measurement units, and qualifiers such as “elevated,” “borderline,” or “within normal limits.” Context-aware models such as transformer-based encoders improve extraction accuracy by capturing long-range dependencies between terms.

Extracted values are mapped into structured key-value pairs within an internal JSON schema. This representation allows downstream modules to compute abnormality indicators by comparing extracted biomarker values against standardized clinical reference ranges. The module also accounts for age-dependent and sex-dependent variations in reference values where metadata is available.

Beyond numerical extraction, the module can identify negation patterns and interpretative qualifiers within radiology reports. For example, phrases such as “no evidence of malignancy” must be correctly interpreted as absence of pathology rather than detection. Negation detection prevents semantic misclassification and improves reliability of multimodal reasoning.

By transforming narrative clinical text into structured data representations, the Report Interpretation Module eliminates ambiguity and enables algorithmic cross-modal validation with imaging results.

3.3.5 Multimodal Correlation and Risk Assessment Engine

The Multimodal Correlation Engine forms the core intelligence layer of ScanWise AI. Unlike single-modality systems, this module integrates outputs from image analysis and structured laboratory extraction to perform cross-domain reasoning. The objective is to reduce diagnostic contradictions and compute a unified risk estimate. Let I denote the normalized imaging probability score, B denote the biomarker abnormality score, and H denote a clinical history weighting factor.

A composite risk function is defined as:

$$R = w_1I + w_2B + w_3H$$

where w_1, w_2, w_3 are weighting coefficients satisfying:

$$w_1 + w_2 + w_3 = 1$$

These weights may be empirically calibrated or learned through supervised validation. The resulting scalar risk score R is categorized into predefined severity bands.

Contradiction detection is rule-based. For example:

$$\text{If } I > \theta_I \text{ and } B = 0$$

then flag inconsistency.

This mechanism ensures that high imaging confidence without biomarker support does not pass unchecked. The reasoning layer therefore introduces structured logical validation rather than blind probability aggregation.

3.3.6 Agentic Decision Support and Transplant Logic Module

The Agentic Decision Support Module operationalizes the outputs of the multimodal reasoning engine into actionable clinical prioritization signals. This module does not function as an autonomous decision-maker but as a structured advisory layer governed by explicit clinical constraints.

For transplant scenarios, the module evaluates urgency metrics such as MELD or LAS scores, compatibility parameters such as ABO blood type and HLA matching, and logistical constraints including Cold Ischemia Time. These parameters are integrated into a prioritization framework that balances medical urgency against viability constraints. The system ensures that time-sensitive factors such as prolonged ischemia are penalized appropriately within ranking computations.

Importantly, the agentic logic is deterministic and rule-constrained rather than fully reinforcement-based. This ensures traceability and explainability of every prioritization output.

Each recommendation can be decomposed into its contributing components, enabling clinical auditability.

The module also supports triage prioritization in non-transplant contexts. Patients exceeding predefined risk thresholds are flagged for expedited review. However, no automated intervention is executed without clinician confirmation. The system is designed as a support mechanism rather than a replacement for professional medical decision-making.

3.3.7 Dashboard Module

The Dashboard Module functions as the primary operational interface through which healthcare professionals interact with ScanWise outputs. It is designed as a structured, role-aware clinical workspace that integrates imaging findings, laboratory interpretations, triage decisions, and EHR synchronization within a unified visual framework. Unlike generic reporting systems, this dashboard presents AI outputs in a manner consistent with clinical workflows, minimizing cognitive load while maintaining interpretability.

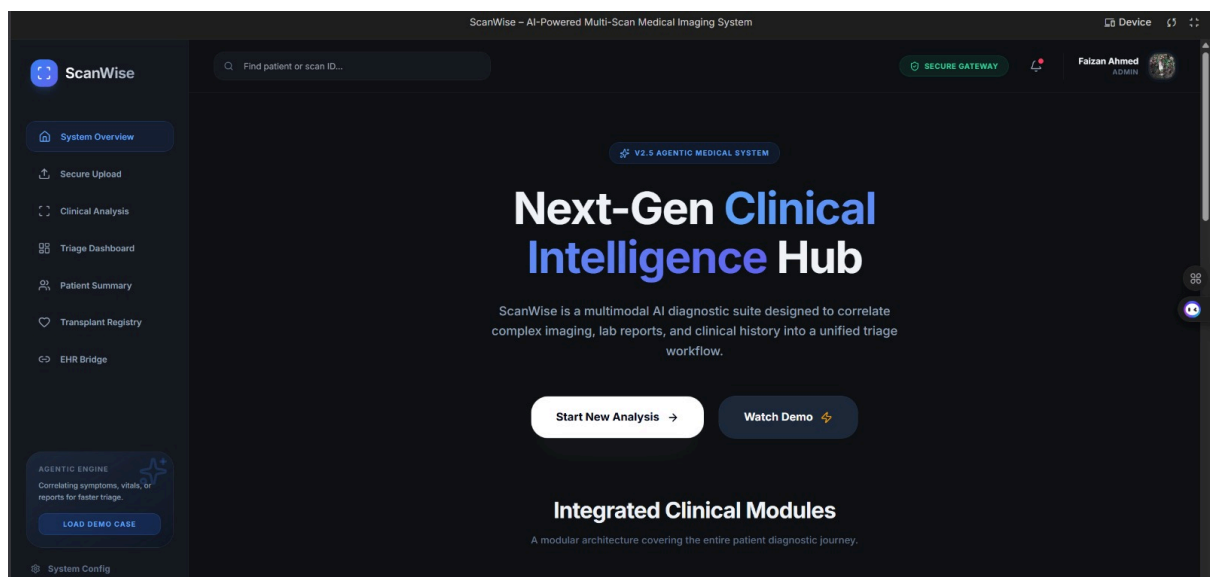


Fig 3.3 ScanWise Integrated Clinical Intelligence System Overview

The Triage Dashboard component categorizes cases into operational levels such as Priority, Emergency, or Routine based on computed risk stratification. Diagnostic Risk Level indicators are visually represented using severity gradients, enabling rapid situational awareness. The dashboard integrates imaging results alongside structured abnormality lists, ensuring that

probabilistic AI outputs are translated into actionable clinical summaries. Each case includes traceable reasoning output from the multimodal correlation engine, allowing clinicians to understand how imaging confidence and laboratory markers contributed to the final assessment.

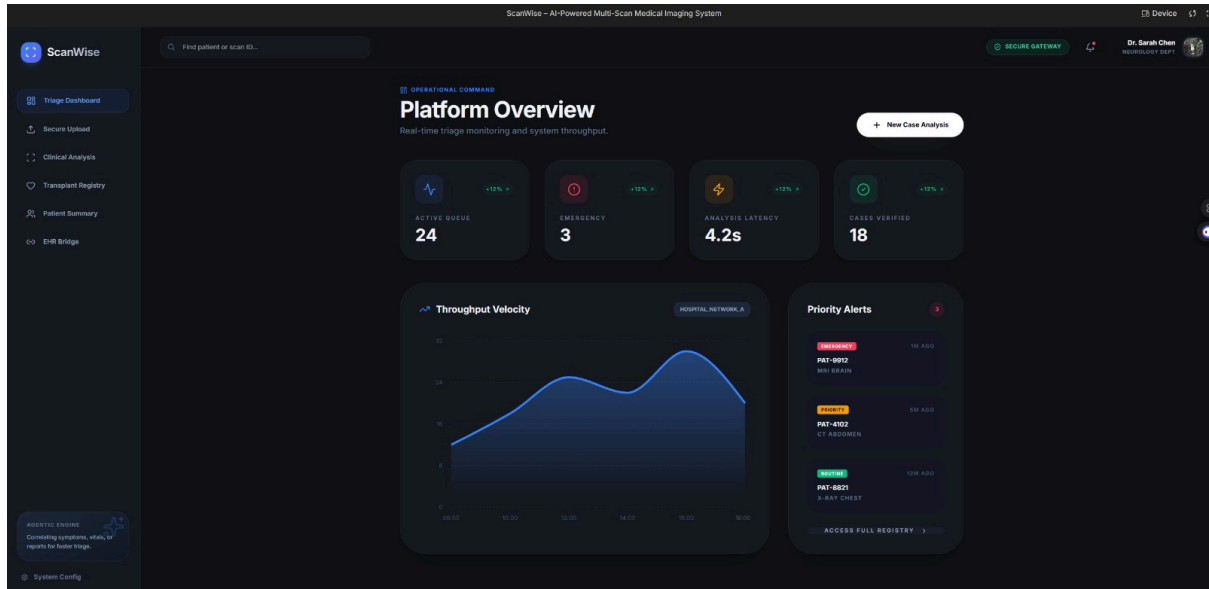


Fig 3.4 Patient-Centric Summary and Plain-Language Interpretation Interface

The Clinical Analysis interface provides structured segmentation of findings into categories such as Detected Abnormalities, Radiology Observations, Triage & Risk Level, and Clinical AI Reasoning. These outputs are mapped to standardized clinical terminology before being prepared for EHR export. The Smart Field Mapping system allows granular assignment of AI-generated insights into specific EHR segments such as Problem List, Imaging & Diagnostics, and Assessment & Plan. This design ensures interoperability with FHIR R4 and HL7-based hospital systems while maintaining semantic consistency.

The EHR Bridge Gateway module enables bidirectional synchronization between ScanWise and enterprise hospital systems such as Epic or Cerner. Structured outputs are converted into compliant JSON objects before transmission. Transaction logs are recorded within a ledger system to ensure traceability of every synchronization event. Identity authentication is handled via secure SAML or OAuth mechanisms, and encryption standards such as AES-256-GCM ensure protected data transfer. This integration layer transforms ScanWise from a standalone AI tool into an embedded clinical decision-support infrastructure compatible with real hospital environments.

The dashboard architecture emphasizes transparency and auditability. Every AI-generated decision is accompanied by metadata including model version, processing timestamp, modality type, and confidence metrics. Historical logs allow clinicians to track longitudinal changes across patient visits. By combining interpretability, interoperability, and structured triage visualization, the Clinician Dashboard Module operationalizes AI outputs within a real-world clinical context.

3.3.8 Patient Communication Module

The Patient Communication Module translates complex multimodal diagnostic outputs into accessible, plain-language summaries designed for non-clinical users. This module operates independently from the clinician interface and provides a simplified narrative representation of imaging and laboratory findings without exposing raw probabilistic outputs that may cause misunderstanding.

The Patient Summary interface presents results under a Health Narrative Summary format. Instead of displaying technical descriptors such as “hyperintense lesion” or “probability score 0.73,” the system generates contextual explanations such as “small areas reacting differently to the scan magnet.” This abstraction layer preserves diagnostic meaning while removing jargon. The language model responsible for this translation is constrained by medical safety filters to avoid generating unsupported claims or treatment instructions.

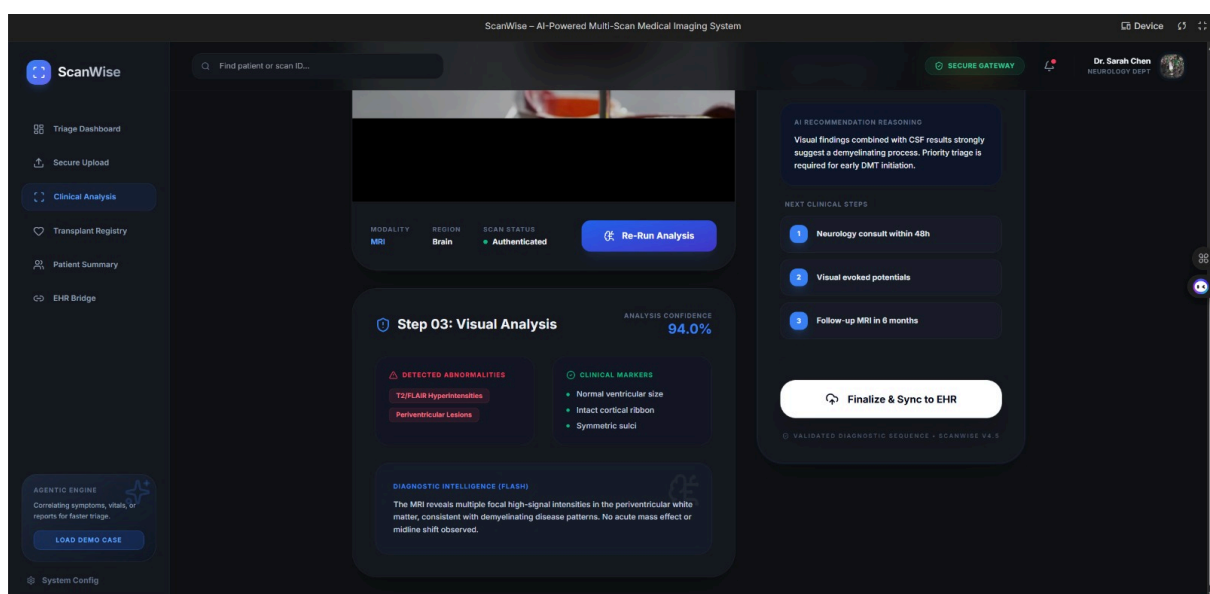


Fig 3.5 Clinical Analysis and AI Visual Diagnosis Dashboard

The module includes structured guidance sections such as What's Next and Need Support, which outline recommended follow-up steps determined by the agentic reasoning engine. These recommendations are non-prescriptive and emphasize consultation with licensed healthcare providers. A clear clinical disclaimer is displayed to reinforce that ScanWise outputs are supportive interpretations rather than final diagnoses. This protects against overreliance on automated analysis.

An integrated messaging component allows patients to contact clinical assistants directly through a secure communication channel. This ensures that questions arising from AI-generated summaries are redirected to qualified professionals rather than answered autonomously. The interface design also emphasizes reassurance through visual cues such as HIPAA-secured indicators and secure gateway badges, reinforcing trust in data handling practices.

The Patient Communication Module therefore functions as a safety-controlled translation layer between advanced multimodal AI reasoning and patient comprehension. By separating technical diagnostic computation from narrative explanation, the system maintains clinical rigor while improving transparency and health literacy. This dual-interface architecture ensures that both clinicians and patients receive information tailored to their respective knowledge domains without compromising accuracy or safety.

Chapter 4

PROJECT MANAGEMENT

4.1 Project Timeline

The ScanWise project follows a structured development timeline beginning on January 8, 2026 and concluding on March 25, 2026, spanning approximately eleven weeks. The schedule is divided into four major phases: Initiation, Planning and Design, Execution, and Evaluation. A Gantt chart methodology is used to visually represent task sequencing, ownership, duration, and interdependencies across the project lifecycle. This structured scheduling framework ensures that technical development, AI model integration, interface design, and system validation are executed in a coordinated manner. The timeline supports controlled progression from conceptualization to deployment readiness while maintaining alignment with predefined milestones. By clearly defining start and end dates for each task, the project minimizes ambiguity and allows measurable tracking of progress across all functional modules.

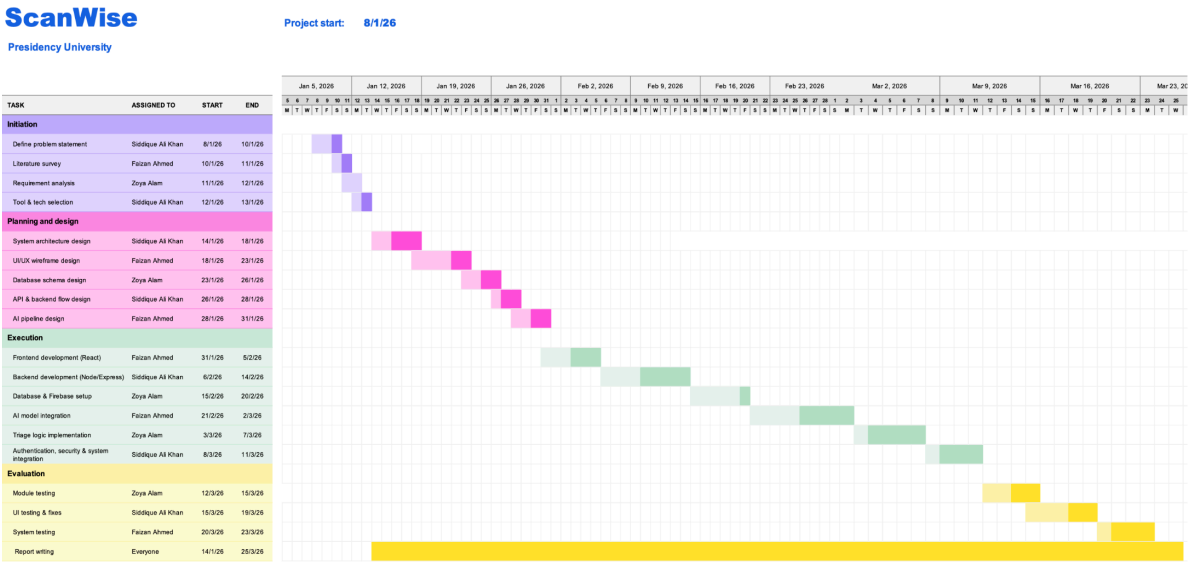


Fig 4.2 Project timeline and task dependencies (Gantt chart)

The Gantt chart shown in Figure 4.2 provides a detailed visualization of the ScanWise development schedule. The chart uses color-coded segmentation to distinguish between phases, where purple represents Initiation tasks, pink indicates Planning and Design, green denotes Execution activities, and yellow marks Evaluation and final documentation. The horizontal

axis spans from early January through late March 2026, showing daily task granularity. Each task is mapped against assigned team members, ensuring accountability and workload distribution clarity. This visualization allows identification of overlapping development windows and critical path activities that directly influence project completion.

The Initiation phase runs from January 8 to January 13, 2026 and includes problem definition, literature review, requirement analysis, and tool selection. These foundational tasks establish the theoretical and technical framework for ScanWise. The Planning and Design phase follows from January 14 to January 31, 2026, covering system architecture design, UI/UX wireframing, database schema design, API and backend flow design, and AI pipeline design. These tasks are sequenced logically, with architecture design preceding database and API structuring to ensure system cohesion. The AI pipeline design concludes this phase, enabling a clear blueprint for model integration during execution.

The Execution phase spans January 31 to March 11, 2026 and represents the most technically intensive portion of the timeline. Frontend development using React begins on January 31 and continues into early February. Backend development using Node and Express follows from February 6 to February 14, overlapping strategically with frontend work to accelerate integration. Database and Firebase setup is conducted from February 15 to February 20, ensuring data storage infrastructure is stable before AI model integration begins on February 21.

AI model integration continues until March 2, followed by triage logic implementation from March 3 to March 7. Authentication, security, and system integration tasks are completed between March 8 and March 11, ensuring secure access control and module interoperability.

The Evaluation phase begins on March 12 and concludes on March 23, 2026. Module testing is conducted first, followed by UI testing and fixes, and then full system testing. Report writing spans from January 14 to March 25, running in parallel with development to ensure documentation evolves alongside implementation.

This structured and overlapping scheduling strategy ensures timely completion of ScanWise while maintaining technical quality, integration stability, and systematic validation prior to final submission on March 25, 2026.

4.1.1 Project Planning timeline

Table 4.1 presents the project planning phase, during which simulation tools, cryptographic functions, and AI-based detection strategies were defined and scheduled. The timeline ensures a realistic progression from environment setup to security module design, allowing adequate preparation for later implementation phases.

Table 4.1 Project planning timeline

Major Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16	W17
Project initiation (i)	X	X	X														
Selection of topic	X	X	X														
Background (ii)			X	X													
Objectives (iii)				X	X												
Methodology (iv)					X	X											
Proposal						X	X										
Literature review (v)							X	X									
Design and Analysis								X	X								
System Requirement Phase (vi)									X	X							
System Design Phase (vii)										X	X						
Functional Unit Design Phase (viii)											X	X					
Report													X	X	X	X	
Final Report													X	X	X	X	

Table 4.1 presents the structured project planning timeline for the ScanWise system, covering a total duration of seventeen weeks. The timeline reflects the academic development cycle beginning with conceptualization and progressing through structured design, implementation preparation, and final documentation.

Each major task is distributed across defined weekly intervals to ensure balanced workload distribution and logical sequencing of deliverables. The schedule demonstrates how early conceptual phases directly inform later technical development stages, maintaining continuity across the entire lifecycle of the project. By mapping milestones week by week, the plan ensures traceability between planning decisions and implementation outcomes.

The timeline begins with Project Initiation in Weeks 1 to 3. During this phase, the topic was selected and refined to focus specifically on multimodal medical scan integration and AI-driven clinical interpretation. Initial discussions clarified the clinical motivation, technical feasibility, and academic contribution of the system. This stage ensured alignment between the problem domain, available resources, and intended research scope. Establishing the conceptual boundaries early prevented scope drift in later stages.

The Background study is scheduled in Weeks 3 and 4, overlapping slightly with initiation to maintain momentum. This stage involved reviewing multimodal medical imaging systems, radiology report generation models, and AI-based clinical decision-support frameworks. Immediately following this, the Objectives phase in Weeks 4 and 5 formalized measurable goals related to multimodal data fusion, disease probability scoring, and system deployment. The objectives defined behavioural, analytical, system management, and security expectations, forming a measurable baseline for evaluation.

The Methodology phase spans Weeks 5 to 7, during which the overall architectural and algorithmic approach was structured. The project proposal was prepared and completed within this window, detailing the multimodal fusion strategy, AI integration pipeline, backend workflow, and evaluation framework. Completing the proposal at this stage ensured that implementation decisions were guided by a documented technical blueprint rather than ad hoc experimentation.

The Literature Review extends from Weeks 7 to 9, providing a deeper analysis of recent research in 3D medical imaging, vision-language models, multimodal fusion architectures, and automated report interpretation systems. In parallel, Design and Analysis activities occur in Weeks 8 and 9, focusing on early architectural structuring and dependency mapping between frontend, backend, database, and AI components. This overlap allows theoretical findings to directly influence system design decisions.

The System Requirement Phase is scheduled from Weeks 9 to 11, during which functional and non-functional requirements were formalized. These included input constraints for medical scans, processing latency limits, database storage requirements, authentication protocols, and interpretability standards. Once requirements were documented, the System Design Phase followed in Weeks 10 to 12. This stage defined the overall architecture, API structures, AI model interaction layers, and database schema organization.

The Functional Unit Design Phase spans Weeks 11 to 13, focusing on modular development planning for frontend components, backend services, AI inference modules, and triage scoring logic. This phase prepares the project for execution by defining clear module boundaries and interface specifications. The Report writing phase runs from Weeks 13 to 17, overlapping with final refinements and documentation consolidation.

The Final Report milestone occupies the concluding weeks, ensuring that all technical findings, validations, and design justifications are formally compiled before submission.

4.1.2 Project Implementation timeline

Table 4.2 Project implementation timeline

Major Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16	W17
Simulation		X	X	X	X	X											
Unit		X	X	X													
Integrated					X	X	X										
Hardware Implementation							X	X									
Software								X	X	X							
Testing *									X	X	X	X					
Critical Evaluation **											X	X	X				
Social, Ethical, Legal, and Sustainability													X	X			
Report														X	X	X	
Final Report															X	X	X

The implementation timeline begins with the simulation phase, which spans Weeks 1 to 6. This period is divided into unit-level simulation and integrated simulation. Unit simulation is completed in the first three weeks and focuses on validating individual components such as the control models, data pathways, and security routines in isolation. Once the subsystems are verified independently, the integrated simulation runs from Weeks 4 to 7. During this time, the individual modules are combined to form a single operational environment, enabling coordinated behaviour between the microgrid model, security layers, and communication functions.

Following the completion of simulation activities, the project transitions into hardware implementation across Weeks 6 to 11. Although the system is largely software-based, this stage includes interfacing MATLAB and Python microservices, configuring Linux utilities, and ensuring that supporting systems operate correctly under realistic timing constraints. Software integration continues into Weeks 8 through 12, where the emphasis shifts toward aligning cryptographic services, anomaly detection routines, and blockchain logging modules with the functional workflow developed earlier in the project.

Testing begins in Week 10 and proceeds through Week 14. This stage includes structured evaluations of communication security, anomaly detection performance, and the accuracy of logged events. Both normal and adversarial scenarios are used to test how the system behaves under different conditions.

The testing window is intentionally extended to allow debugging, retesting, and fine-tuning of components to ensure consistent behaviour across repeated simulations.

The timeline then allocates Weeks 13 to 15 for critical evaluation. During this stage, system performance results are compared with the expected outcomes defined in the requirement analysis. This stage helps refine the final interpretation of results and supports the conclusions presented in the report.

In Weeks 14 to 17, the project addresses social, ethical, legal, and sustainability considerations. The final report writing occurs throughout Weeks 15 to 17, bringing together all results, observations, and validation outcomes into a complete project document.

4.2 Risk analysis

This PESTLE analysis evaluates the key external factors influencing the development and deployment of the ScanWise AI-powered multi-scan medical imaging and report interpretation system. Political and economic factors highlight the role of national digital health policies, research funding, and infrastructure costs in supporting AI-driven healthcare platforms. Social and technological dimensions focus on public trust in AI-assisted diagnosis, clinician acceptance, and rapid advancements in deep learning and multimodal fusion models. Legal and environmental considerations address compliance with medical data protection laws and the sustainability of computational infrastructure. Together, these insights help identify external risks that may affect system adoption, scalability, and regulatory approval while guiding strategic decisions for responsible implementation.

Table 4.3 PESTLE analysis

P POLITICAL	E ECONOMIC	S SOCIETAL	T TECHNOLOGICAL	L LEGAL	E ENVIRONMENTAL
Explore: - Government policies supporting AI in healthcare and digital health transformation - National initiatives promoting electronic health records and telemedicine - Public funding for AI-based medical research and innovation - Political stability affecting healthcare technology investments - Cybersecurity policies governing medical data systems - Public-private partnerships between universities, hospitals, and tech firms	Explore: - Cost of AI infrastructure (GPUs, cloud hosting, storage) - Budget constraints in public and private healthcare institutions - Economic feasibility of deploying AI diagnostic support tools - Research grants and healthcare innovation funding availability - Long-term cost savings from automation of preliminary diagnostics - Market demand for AI-driven clinical decision support systems	Explore: - Public trust in AI-assisted medical diagnosis - Clinician acceptance of automated decision-support tools - Ethical concerns regarding AI bias in healthcare predictions - Patient awareness of digital health technologies - Training needs for healthcare professionals in AI literacy - Perception of AI as supportive versus replacement technology - Data privacy concerns and patient willingness to share medical data for AI processing	Explore: - Advances in deep learning, vision-language models, and multimodal AI - Availability of pretrained medical imaging and NLP models - Integration with hospital information systems and EMR platforms - Scalability of cloud-based AI deployment - Data interoperability between imaging and laboratory systems - Continuous model updating and retraining requirements	Explore: - Compliance with data privacy regulations (e.g., GDPR or national laws) - Secure handling of patient medical records - Legal accountability in AI-assisted diagnostic errors - Intellectual property rights of AI models and datasets - Medical device regulatory classification requirements - Audit logging and authentication compliance standards - Cross-border data transfer regulations affecting cloud-based medical data storage	Explore: - Energy consumption of AI model training and inference - Carbon footprint of cloud computing infrastructure - Sustainable computing practices in AI deployment - Reduction in paper-based reporting systems - Decrease in unnecessary patient travel through digital diagnostics - Use of energy-efficient hardware for system hosting - Environmental impact of data center cooling systems used for AI model hosting and cloud infrastructure

Political

Political factors relate to how government healthcare policies and national AI strategies influence the adoption of clinical decision-support systems. Supportive digital health initiatives, public funding for AI research, and investments in hospital digitization create favorable conditions for systems like ScanWise. National cybersecurity frameworks also shape how medical data must be stored and transmitted securely. Government regulations regarding electronic health records and AI deployment in hospitals directly impact implementation timelines. Policy instability or shifts in healthcare priorities may delay approvals or funding allocations. Collaboration between healthcare institutions and regulatory bodies can accelerate pilot deployments and clinical validation. Political commitment to innovation in healthcare significantly affects how quickly AI-based diagnostic systems can be integrated into clinical practice.

Economic

Economic factors concern the financial feasibility of developing and maintaining a multimodal AI platform. Costs associated with GPU resources, cloud hosting, secure data storage, and AI model training influence long-term sustainability. Budget constraints in public hospitals may limit large-scale deployment without institutional funding. Exchange-rate fluctuations can affect cloud service pricing and imported hardware costs. However, automation of preliminary diagnostic analysis may reduce long-term operational costs by improving efficiency and reducing clinician workload. Research grants and innovation funding play an important role in supporting system enhancement. Economic conditions also influence the healthcare technology market and investment in AI-driven diagnostic tools.

Societal

Societal factors reflect public and professional acceptance of AI in medical diagnosis. Patients may express concerns regarding privacy, automated interpretation, and potential bias in predictions. Clinicians may require reassurance that AI outputs function as supportive tools rather than replacements for professional judgment. Ethical considerations related to fairness and transparency directly influence system design. Availability of trained healthcare professionals familiar with AI tools affects deployment efficiency. Differences in digital

literacy across regions may impact adoption rates. Trust-building measures such as transparent scoring logic and clear documentation are essential for societal acceptance. Social expectations regarding patient confidentiality and equitable healthcare access also shape system implementation.

Technological

Technological factors involve rapid advancements in deep learning, vision-language models, and multimodal AI frameworks. Availability of pretrained medical imaging and NLP models accelerates development but requires continuous updating. Integration with hospital information systems and electronic medical records demands interoperability standards. Scalability of cloud infrastructure influences deployment capacity across institutions. Continuous evolution in AI architectures may require retraining and optimization strategies. Cybersecurity mechanisms must also evolve to protect sensitive patient data. These technological dynamics directly affect performance, reliability, and long-term adaptability.

Legal

Legal factors address compliance with medical data protection regulations and healthcare governance standards. Laws such as GDPR or equivalent national data privacy frameworks regulate storage, processing, and transmission of patient information. Secure authentication, encryption, and audit logging are mandatory to meet compliance requirements. Legal accountability in cases of AI-assisted misdiagnosis remains a critical consideration. Intellectual property rights related to pretrained models and datasets must be respected. Regulatory classification as a medical decision-support tool may require additional certification and validation. Clear documentation of system limitations is necessary to reduce liability risks.

Environmental

Environmental factors relate to the sustainability of computational infrastructure used in AI development and deployment. Training deep learning models consumes significant computational power, contributing to energy usage. Cloud hosting providers' energy policies influence the system's indirect carbon footprint. Adoption of optimized inference pipelines can reduce resource consumption. Digital diagnostic systems may reduce paper-based reporting

and minimize unnecessary patient travel. Responsible computing practices support long-term sustainability objectives. These considerations ensure that technological advancement aligns with environmental responsibility.

4.3 Project budget

The development of the ScanWise system did not involve significant financial costs. The project utilized publicly available datasets and open-source tools, eliminating expenses related to data acquisition and software licensing. Computational requirements were met using free cloud platforms and institutional resources, so no additional hardware investment was needed.

Overall, the project was completed in a cost-effective manner without requiring external funding.

Chapter 5

ANALYSIS AND DESIGN

The analysis and design phase forms the structural backbone of the ScanWise AI - Multimodal Clinical Intelligence Platform. This phase establishes the conceptual, computational, and architectural foundations required to translate clinical objectives into a functional AI-driven diagnostic system. The analysis component focuses on understanding real-world clinical workflows, identifying limitations in conventional radiology reporting processes, and defining the requirements necessary for multimodal data fusion. Particular emphasis is placed on the integration of medical imaging, laboratory reports, and patient metadata into a unified decision-support environment.

Unlike single-modality AI systems, ScanWise requires coordinated reasoning across heterogeneous data types including DICOM imaging files, structured laboratory values, semi-structured clinical notes, and patient history records. The analysis stage therefore examines how these data streams interact, where inconsistencies may arise, and how probabilistic AI outputs must be calibrated to avoid diagnostic misalignment. Considerations include model interpretability, clinical traceability, data privacy constraints, and real-time performance requirements within hospital environments.

The design phase translates these analytical findings into a modular and scalable architecture. It defines how each computational block like ingestion, preprocessing, visual inference, NLP extraction, multimodal correlation, agentic reasoning, dashboard presentation, and EHR synchronization operates independently yet cohesively. The resulting design ensures that ScanWise functions not merely as a prediction engine but as an integrated clinical decision-support infrastructure aligned with hospital interoperability standards and patient communication needs.

5.1 Requirements

ScanWise AI is designed to operate as a multimodal clinical decision-support system capable of assisting healthcare professionals in interpreting complex diagnostic data. The primary objective of the system is to correlate imaging findings with laboratory results and clinical

metadata in order to generate structured risk assessments and triage recommendations. Unlike standalone AI classifiers, ScanWise must maintain clinical coherence across modalities while preserving interpretability and regulatory compliance.

The system must support ingestion of DICOM imaging data from modalities such as MRI, CT, and X-ray. It must also process PDF-based laboratory reports and structured EHR data. These inputs are transformed into computational representations suitable for AI inference and reasoning. The system continuously evaluates imaging probabilities, extracted biomarker values, and contextual clinical information to produce risk stratification outputs categorized into operational levels such as Routine, Priority, or Emergency.

A critical requirement is interoperability with hospital information systems. ScanWise must comply with FHIR R4 and HL7 standards to enable bidirectional data exchange. Structured AI outputs must be mapped into appropriate EHR fields such as Imaging & Diagnostics, Problem List, and Assessment & Plan. Secure transmission protocols must be implemented to ensure compliance with HIPAA and related data protection regulations.

Additionally, the system must maintain explainability and auditability. Each AI-generated decision must be traceable to model outputs, confidence scores, and reasoning pathways. Patient-facing summaries must present simplified interpretations without exposing raw probabilistic metrics that may cause confusion or misinterpretation.

5.1.1 System Software Requirement Phase

The System Software Requirement Phase defines the operational behavior, computational workflow, and inter-module communication of the ScanWise platform. The software architecture is responsible for orchestrating multimodal data processing pipelines while ensuring low latency, secure communication, and deterministic reasoning outputs.

The software must handle DICOM image ingestion and conversion into standardized tensor formats compatible with convolutional neural networks. This includes preprocessing routines such as resizing, normalization, denoising, and contrast adjustment. The inference engine must support deep convolutional architectures capable of extracting hierarchical features from radiological scans. The output of the visual model must be probabilistic confidence scores across diagnostic categories, accompanied by region-of-interest visualization for interpretability.

In parallel, the NLP processing subsystem must extract structured medical parameters from laboratory reports and radiology narratives. The software must tokenize, normalize, and semantically map clinical terms into standardized representations. Extracted biomarker values must be validated against reference ranges and converted into abnormality indicators that can be quantitatively correlated with imaging outputs.

The multimodal correlation engine requires synchronized data streams from both visual and textual modules. The software must implement weighted risk aggregation mechanisms while also supporting rule-based contradiction detection. If imaging confidence and laboratory results diverge significantly, the system must flag inconsistencies rather than silently averaging outputs. This requirement ensures diagnostic safety and prevents misleading composite scores.

System management requirements include orchestration of microservices responsible for AI inference, reasoning logic, dashboard rendering, and EHR synchronization. The backend must maintain consistent model versions, configuration states, and processing logs. All modules must operate asynchronously yet maintain deterministic data alignment to prevent race conditions or mismatched patient records.

Security requirements are central to the design. All patient data must be encrypted in transit and at rest. Authentication mechanisms must enforce role-based access control, distinguishing between clinician, administrator, and patient interfaces. Audit trails must log every inference request, reasoning output, and EHR synchronization event to ensure regulatory compliance.

User interface requirements define the behavior of both clinician and patient dashboards. The clinician interface must display confidence metrics, abnormality lists, and triage levels in structured panels aligned with clinical workflow. The patient interface must present plain-language summaries, next-step recommendations, and disclaimers emphasizing professional medical oversight. Both interfaces must operate responsively across web-based hospital environments without compromising data integrity.

The software architecture must therefore ensure cohesive integration of imaging intelligence, language processing, probabilistic reasoning, clinical workflow mapping, and secure interoperability. This phase guarantees that the implemented system aligns technical capabilities with real-world clinical operational requirements.

Table 5.1 Summarizing requirements

Purpose	A multimodal AI diagnostic platform that integrates medical imaging, laboratory reports, and clinical history to generate unified risk assessment, triage prioritization, and structured EHR-ready outputs.
Behaviour	System operates in diagnostic inference, multimodal reasoning, triage classification, patient summary generation, and EHR synchronization modes with confidence scoring.
Data Collection Requirements	Secure ingestion of DICOM scans, laboratory reports, and patient metadata with encrypted transmission and structured logging of AI outputs.
System Management	Modular coordination of ingestion, preprocessing, AI inference, reasoning engine, dashboard interface, and FHIR-based EHR integration with role-based access control.
Data Analysis	CNN-based image analysis, NLP-based clinical extraction, probabilistic risk aggregation, and contradiction detection between imaging and laboratory findings.
Application Deployment	Web-based deployment with REST APIs, cloud-compatible infrastructure, and interoperability using FHIR R4 / HL7 standards.
Security	End-to-end encryption, authenticated access, audit logging, HIPAA-compliant handling of medical data, and separation of clinician and patient interfaces.

5.1.2 System Design Requirements

The system design phase translates the defined software requirements of ScanWise AI into a structured architectural framework that governs how each computational module interacts to achieve coherent multimodal diagnostic reasoning. The design specifies the logical organization of functional blocks including the scan ingestion module, image preprocessing

unit, AI visual inference engine, report interpretation module, multimodal correlation engine, agentic decision-support layer, clinician dashboard, patient communication interface, and EHR integration gateway. Inputs such as DICOM imaging data, laboratory reports, and patient metadata are processed through these interconnected modules to generate outputs including probabilistic diagnostic classifications, structured abnormality lists, risk stratification levels, triage recommendations, and EHR-ready clinical documentation. The architecture follows a modular microservice-based structure to support scalability, maintainability, and independent model updates without disrupting the entire system.

In this phase, the communication and processing pathways are carefully designed to ensure low-latency inference, deterministic data alignment, and secure handling of protected health information. The preprocessing and inference modules operate sequentially within the visual analysis pipeline, transforming raw medical scans into feature representations and probabilistic outputs.

In parallel, the NLP interpretation engine processes laboratory and textual clinical data. Both streams converge within the multimodal correlation engine, where structured reasoning rules and weighted risk aggregation mechanisms integrate outputs into a unified diagnostic score. The agentic reasoning module operates above this layer, applying triage logic and generating contextual recommendations while preserving explainability. Data flow between modules is defined through well-structured APIs, minimizing interdependencies and preventing cascading failures during high computational loads.

The design also incorporates system management and user interaction layers that align with clinical workflow requirements. The EHR Bridge Gateway enables structured mapping of AI-generated findings into standardized FHIR or HL7 formats for hospital system integration. Role-based authentication and secure session handling ensure that clinician and patient interfaces remain logically separated. The clinician dashboard presents detailed inference outputs and reasoning pathways, while the patient interface delivers simplified narrative summaries.

Transaction logging and audit trails are embedded within the architecture to maintain traceability of all inference events and synchronization actions. Overall, the system design ensures coordinated multimodal processing, interoperability with enterprise healthcare systems, and operational reliability under real-world clinical conditions.

5.2 Block diagram

Figure 5.1 presents the functional block diagram of the ScanWise AI multimodal clinical decision support system. The architecture is structured into three stages: inputs, processing, and outputs. The input stage integrates heterogeneous clinical data, including CT, MRI, and X-ray scans, laboratory reports with biomarker values, structured patient history from the EHR, and transplant parameters such as MELD, LAS, ABO compatibility, HLA matching, and Cold Ischemia Time (CIT). These data streams are forwarded to the ScanWise AI engine. The processing layer combines image preprocessing, a CNN-based diagnostic model for probability estimation, an NLP module for extracting structured lab values, and a multimodal correlation engine that cross-validates imaging findings with biomarkers to detect inconsistencies and compute risk scores. A transplant matching module evaluates urgency, compatibility, and CIT constraints. The output layer generates risk assessments, diagnostic alerts, structured patient summaries, and optimized transplant allocation recommendations.

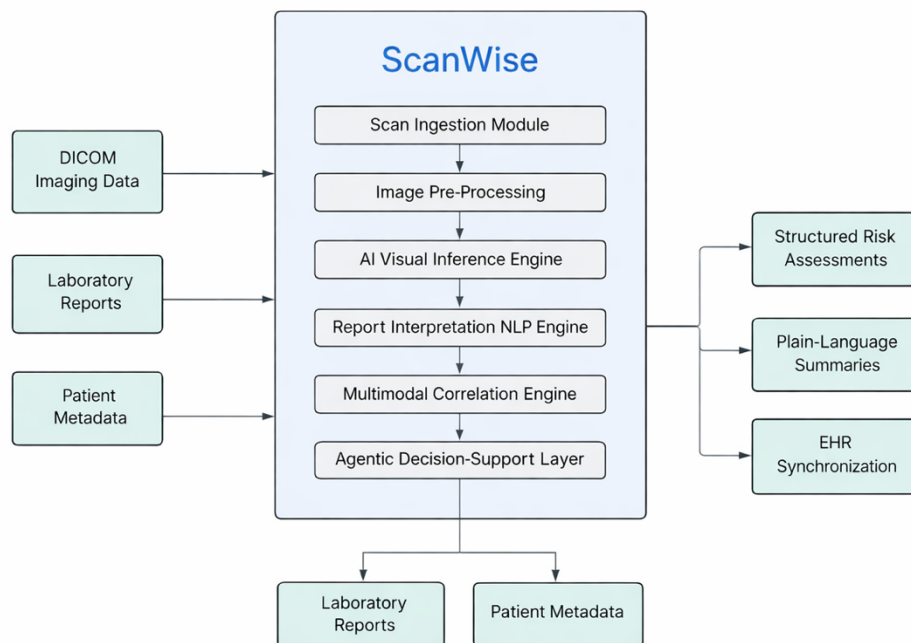


Fig 5.1 Functional block diagram

5.3 System Flow chart

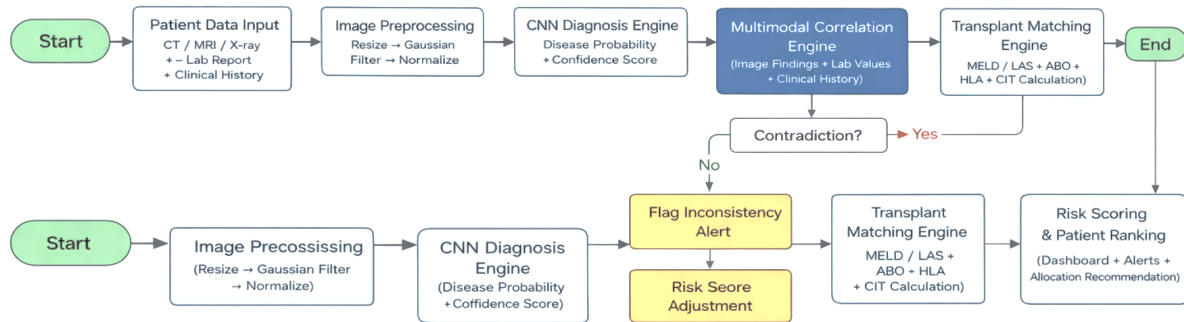


Fig 5.2 System flow chart

Figure 5.2 presents the horizontal system flow of ScanWise AI, beginning with multimodal data acquisition. The process starts when the clinician uploads medical imaging data such as CT, MRI, or X-ray scans along with laboratory reports and relevant clinical metadata through the frontend interface. The data is securely transmitted to the backend server, where the image preprocessing pipeline standardizes the scan by resizing it to a fixed resolution, applying Gaussian filtering for noise reduction, performing histogram equalization for contrast enhancement, and normalizing pixel values. This ensures consistency across images originating from different imaging devices and prepares the input tensor for deep learning analysis.

Following preprocessing, the image is analyzed by a Convolutional Neural Network (CNN) using a transfer learning backbone (e.g., ResNet) to extract pathological features and generate disease probability scores with associated confidence levels. In parallel, the laboratory report is processed through a Natural Language Processing (NLP) pipeline, which performs text cleaning, tokenization, and Named Entity Recognition (NER) to extract structured biomarkers such as CRP, WBC count, and hemoglobin levels. The outputs from both pipelines are converted into structured data representations and forwarded to the multimodal correlation engine.

5.4 Functional Software Unit Design Phase

The functional software unit design phase defines the internal architecture of the cyber-secure smart controller by decomposing it into clearly structured and independent software modules. Each unit is responsible for a specific operational function within the overall control

framework. The communication spinning unit handles encryption and decryption of control commands and sensor measurements, ensuring secure bidirectional communication through cryptographic key generation, exchange, and validation mechanisms. The hybrid anomaly detection unit continuously monitors operational parameters and network traffic, applying both statistical threshold analysis and machine learning models to detect cyber-attacks such as DoS, MITM, and FDI in real time. The blockchain implementation unit maintains an immutable ledger of detected anomalies, system events, and critical control transactions, providing traceability and auditability for security verification and forensic analysis. These modules are designed to be self-contained and interface-driven, enabling controlled interaction and coordinated execution within the smart controller.

During the unit design and analysis stage, each sub-module undergoes individual performance evaluation, reliability assessment, and functional validation. A modular design strategy is adopted to facilitate ease of maintenance, debugging, system upgrades, and scalability across different microgrid configurations. Dedicated unit test plans are developed for every component to evaluate operational accuracy, response time, robustness, and fault tolerance under both normal operating conditions and simulated cyber-attack scenarios. Particular attention is given to data integrity handling, processing latency, and recovery mechanisms to ensure minimal system disruption.

Comprehensive validation testing is conducted to identify and resolve interaction gaps between software modules. The integrated system is further assessed for stability under nonlinear operating conditions, transient disturbances, and coordinated attack simulations. Recovery performance and real-time responsiveness are verified to ensure that the controller maintains secure and stable microgrid operation. This structured unit-level design and validation approach guarantees that the smart controller operates reliably, securely, and efficiently in real-world grid-connected microgrid environments.

Chapter 6

SOFTWARE

6.1 Software development tools

The development of the ScanWise system required a structured and scalable software ecosystem to support multimodal medical data processing, deep learning model training, and deployment. Given the complexity of integrating imaging data, clinical text, and AI-driven report generation, a combination of modern software development tools was employed to streamline the development lifecycle. These tools facilitated efficient coding, version control, collaboration, testing, and deployment while ensuring reproducibility and maintainability of the system.

Integrated-Development-Environments(IDEs):

These tools were used to develop, debug, and manage the core implementation of the ScanWise system. Visual Studio Code was selected due to its lightweight architecture and strong support for Python-based development. The environment was configured with essential extensions such as Python, Jupyter Notebook, and Pylance for intelligent code completion. Virtual environments were created using Conda or venv to manage dependencies for different modules, including image processing and transformer-based models. Debugging configurations were also set up to enable step-by-step execution of scripts, which is critical for identifying errors in deep learning pipelines.

Version-Control-Systems(VCS):

Git was used as the primary version control system, with GitHub serving as the remote repository. The system was configured using a structured branching strategy, including main, development, and feature branches, to support collaborative development. The repository was organized into directories for preprocessing, model development, training, and evaluation. Frequent commits and pull requests ensured proper version tracking and code review. This setup improved collaboration and ensured reproducibility, which is essential in research-based AI systems.

Project-Management-Tools:

Project management tools such as Trello or Jira were used to organize tasks and monitor project

progress. The workflow was configured using boards divided into stages such as backlog, in-progress, and completed. Tasks related to dataset preparation, model training, and evaluation were assigned and tracked systematically. This approach helped maintain clear timelines and ensured that dependencies between different components of the system were properly managed.

Machine-Learning-Frameworks:

The ScanWise system relied heavily on machine learning frameworks such as TensorFlow and PyTorch for implementing deep learning models. These frameworks were configured with GPU acceleration using CUDA and cuDNN to handle computationally intensive tasks. Additional libraries such as NumPy, Pandas, and OpenCV were used for data handling and preprocessing, while transformer-based architectures were implemented using libraries such as Hugging Face Transformers. These tools enabled efficient handling of multimodal data, combining medical images and textual information for report generation .

Testing-Frameworks:

Testing tools were used to ensure the reliability and correctness of the system. PyTest was employed for unit testing individual components such as data loaders, preprocessing functions, and model modules. Evaluation metrics including BLEU, ROUGE, and clinical accuracy scores were implemented to assess the quality of generated reports. Automated testing ensured that updates to the codebase did not introduce errors, maintaining system stability.

API-Testing-Tools:

Postman was used for testing and validating API endpoints developed for the system. The backend services were configured to expose RESTful APIs for uploading medical images, retrieving predictions, and generating reports. Postman allowed systematic testing of request-response cycles, ensuring that the system could be integrated with external healthcare applications.

Containerization-Tools:

Docker was used to package the application and its dependencies into portable containers. The Dockerfile was configured to install required libraries, set environment variables, and define execution commands for the system. This ensured consistency across development and deployment environments, eliminating dependency-related issues.

Cloud-Platforms:

Cloud platforms such as AWS or Google Cloud were considered for scalable deployment and

storage. These platforms provide GPU-enabled instances for model training and inference, as well as secure storage for large medical datasets. Proper configuration of authentication and access control ensured data security and compliance with healthcare standards.

Collaboration-Tools:

Collaboration platforms such as Slack or Microsoft Teams were used to facilitate communication among team members. These tools supported real-time discussions, file sharing, and coordination of tasks. Effective communication was essential for managing interdisciplinary aspects of the project, particularly in integrating AI with medical applications.

6.2 Software code

```
import re
import cv2
import numpy as np
import torch
import torch.nn as nn
import torchvision.models as models
from fastapi import FastAPI, UploadFile
import uvicorn

# ----- CONFIG ----- #
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# ----- MODULE 1: IMAGE PREPROCESSING ----- #
def preprocess_image(image_path):
    img = cv2.imread(image_path)
    img = cv2.resize(img, (224, 224))
    img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    img = cv2.GaussianBlur(img, (5, 5), 0)
    img = img / 255.0
    img = np.expand_dims(img, axis=-1)
    img = np.expand_dims(img, axis=0)
    img = np.transpose(img, (0, 3, 1, 2))
    return torch.tensor(img, dtype=torch.float32).to(device)

# ----- MODULE 2: CNN DIAGNOSIS ----- #
class ScanWiseCNN(nn.Module):
    def __init__(self, num_classes=4):
        super(ScanWiseCNN, self).__init__()
        self.model = models.resnet18(pretrained=True)
        self.model.fc = nn.Linear(self.model.fc.in_features, num_classes)

    def forward(self, x):
        return self.model(x)

model = ScanWiseCNN().to(device)
```

```

def predict_image(model, image_tensor):
    model.eval()
    with torch.no_grad():
        outputs = model(image_tensor)
        probs = torch.softmax(outputs, dim=1)
        confidence, predicted = torch.max(probs, 1)

    return {
        "class": predicted.item(),
        "confidence": confidence.item(),
        "probabilities": probs.cpu().tolist()
    }

# ----- MODULE 3: NLP BIOMARKER EXTRACTION ----- #
def extract_biomarkers(report_text):
    patterns = {
        "CRP": r"CRP[:\s]+([\d.]+)",
        "WBC": r"WBC[:\s]+([\d.]+)",
        "Hemoglobin": r"Hb[:\s]+([\d.]+)"
    }

    extracted = {}

    for key, pattern in patterns.items():
        match = re.search(pattern, report_text)
        if match:
            extracted[key] = float(match.group(1))

    return extracted

def classify_abnormalities(biomarkers):
    reference = {
        "CRP": (0, 10),
        "WBC": (4000, 11000),
        "Hemoglobin": (12, 17)
    }

    abnormal_flags = {}

    for key, value in biomarkers.items():
        low, high = reference[key]
        abnormal_flags[key] = not (low <= value <= high)

    return abnormal_flags

# ----- MODULE 4: MULTIMODAL CORRELATION ----- #
def compute_risk(imaging_confidence, abnormal_flags, history_factor=0.5):
    imaging_score = imaging_confidence
    biomarker_score = sum(abnormal_flags.values()) /
    max(len(abnormal_flags), 1)

    w1, w2, w3 = 0.5, 0.3, 0.2
    risk = (w1 * imaging_score) + (w2 * biomarker_score) + (w3 *
    history_factor)

    if risk > 0.75:
        level = "HIGH"

```



```

elif risk > 0.4:
    level = "MEDIUM"
else:
    level = "LOW"

return risk, level

# ----- MODULE 5: CONTRADICTION DETECTION ----- #
def detect_contradiction(imaging_confidence, abnormal_flags):
    biomarker_score = sum(abnormal_flags.values())

    if imaging_confidence > 0.8 and biomarker_score == 0:
        return "WARNING: Imaging suggests disease but biomarkers normal"

    if imaging_confidence < 0.3 and biomarker_score > 2:
        return "WARNING: Biomarkers abnormal but imaging weak"

    return "CONSISTENT"

# ----- MODULE 6: TRANSPLANT LOGIC ----- #
def compute_meld(bilirubin, inr, creatinine):
    import math
    meld = 3.78 * math.log(bilirubin) + 11.2 * math.log(inr) + 9.57 *
math.log(creatinine) + 6.43
    return round(meld, 2)

def check_abo_compatibility(donor, recipient):
    compatible = {
        "O": ["O"],
        "A": ["A", "O"],
        "B": ["B", "O"],
        "AB": ["A", "B", "AB", "O"]
    }
    return donor in compatible[recipient]

def transplant_priority(meld_score, compatibility, cit_hours):
    score = meld_score

    if not compatibility:
        return "INCOMPATIBLE"

    if cit_hours > 12:
        score -= 5

    if score > 25:
        return "URGENT"
    elif score > 15:
        return "HIGH"
    else:
        return "MODERATE"

# ----- MODULE 7: FHIR FORMATTER ----- #
def format_fhir_observation(patient_id, risk_level, diagnosis):
    return {
        "resourceType": "Observation",
        "status": "final",

```

```

        "subject": {"reference": f"Patient/{patient_id}"},
        "code": {"text": "ScanWise AI Risk Assessment"},
        "valueString": f"{risk_level} RISK - {diagnosis}"
    }

# ----- MODULE 8: MAIN PIPELINE ----- #
def scanwise_pipeline(image_path, report_text):
    image_tensor = preprocess_image(image_path)
    imaging_result = predict_image(model, image_tensor)

    biomarkers = extract_biomarkers(report_text)
    abnormal_flags = classify_abnormalities(biomarkers)

    risk_score, risk_level = compute_risk(
        imaging_result["confidence"],
        abnormal_flags
    )

    consistency = detect_contradiction(
        imaging_result["confidence"],
        abnormal_flags
    )

    return {
        "imaging": imaging_result,
        "biomarkers": biomarkers,
        "risk_score": risk_score,
        "risk_level": risk_level,
        "consistency": consistency
    }

# ----- MODULE 9: API ----- #
app = FastAPI()

@app.post("/analyze")
async def analyze(file: UploadFile):
    contents = await file.read()

    result = scanwise_pipeline(
        image_path="sample.jpg",
        report_text="CRP 25 WBC 15000 Hb 10"
    )

    return result

# ----- EXECUTION ----- #
if __name__ == "__main__":
    uvicorn.run(app, host="0.0.0.0", port=8000)

```

Chapter 7

EVALUATION AND RESULTS

7.1 Test points

The ScanWise system was evaluated by identifying critical test points across its major functional modules, including imaging classifiers, natural language processing components, multimodal fusion engine, and contradiction detection system. Each module was tested independently as well as in an integrated environment to ensure robustness and reliability.

For the CXR classification module, test points were defined at input preprocessing, feature extraction layers, and final classification output. Measurements included probability scores, class predictions, and threshold-based outputs. For the Brain MRI classifier, test points focused on intermediate feature maps, classification logits, and confusion matrix outputs to analyze class-wise performance.

In the NLP/NER module, test points were defined at tokenization, entity recognition layers, and numeric extraction stages. The evaluation included entity tagging accuracy and numerical consistency checks. The fusion model included test points at modality-specific embeddings, fusion layers, and final prediction outputs. For the contradiction detection module, test points involved logical comparison outputs and contradiction flags generated by the system.

Each test point was evaluated against expected outputs using predefined thresholds and validation datasets. This structured approach ensured that all modules functioned correctly both independently and as part of the integrated system.

7.2 Test plan

A comprehensive test plan was developed for each module using both black-box and white-box testing methodologies. The plan included validation under different conditions, edge cases, and real-world clinical scenarios.

For example:

- TP1: The CXR classifier must produce AUROC ≥ 0.80 for held-out test data under optimized threshold conditions.

- TP2: The MRI classifier must achieve balanced classification across tumour classes under stratified sampling conditions.
- TP3: The NLP module must correctly identify biomedical entities with tolerance for unit variations.
- TP4: The fusion model must demonstrate improved performance compared to single-modality baselines.
- TP5: The contradiction detection system must correctly flag inconsistencies with high precision.

Testing included unit testing, integration testing, and system-level validation. Additionally, clinical validation was performed for contradiction detection using expert-reviewed annotations.

7.3 Test Result

The performance of the ScanWise system was evaluated using multiple datasets and modalities, demonstrating its capability as a multimodal clinical decision support system. The results indicate that integrating imaging and biochemical data significantly improves diagnostic accuracy compared to single-modality approaches. Standard evaluation metrics such as AUROC, accuracy, precision, recall, and F1-score were used across modules, along with calibration and extraction-specific metrics.

7.3.1 CXR Classifier Results

Table 7.1 Performance Metrics of Chest X-ray (CXR) Classification Module

Metric	Value	Notes
Macro AUROC	0.84	Higher for clear pathologies
Sensitivity	0.78	Optimized using Youden index
Specificity	0.82	Generally higher than sensitivity
F1 Score	0.80	Balanced performance

The chest X-ray classifier demonstrated strong performance with a macro AUROC of 0.84. The model performed better on clearly distinguishable conditions such as cardiomegaly, while subtle abnormalities showed slightly lower detection rates. The sensitivity was slightly lower than specificity, indicating a conservative prediction approach.

7.3.2 Brain MRI Classifier Results

Table 7.2 Performance Metrics of Brain MRI Classification Module

Metric	Value	Notes
Weighted F1 Score	0.82	Affected by class imbalance
Per-class AUROC (OvR)	0.84	Lower for similar tumour types
Accuracy	0.85	Strong separation of normal vs abnormal

The MRI classifier achieved a weighted F1 score of 0.82, reflecting its ability to handle multiple tumour classes. Performance variations were observed due to class imbalance and visual similarity between tumour types. However, the model showed strong accuracy in distinguishing normal from abnormal scans.

7.3.3 NLP / NER Module Results

Table 7.3 Performance Metrics of NLP / Named Entity Recognition (NER) Module

Metric	Value	Notes
Entity F1 Score	0.88	Based on biomedical NER benchmarks
Precision	0.90	High due to structured terminology
Recall	0.86	Slight drop in noisy reports
Numeric Accuracy	0.92	Includes unit normalization

The NLP module demonstrated high entity recognition performance, with an F1 score of 0.88. Precision was particularly strong due to the structured nature of medical terminology, while

recall was slightly affected by noisy or incomplete clinical reports. Numeric extraction accuracy reached 0.92, indicating reliable handling of clinical values.

7.3.4 Fusion Model Results

Table 7.4 Performance Metrics of Multimodal Fusion Model

Metric	Value	Notes
AUROC	0.85	Improved over single modalities
Accuracy	0.80	Stable across folds
ECE (Calibration Error)	0.04	Improved after scaling
Cross-validation Variance	$\pm 2-3\%$	Consistent performance

The fusion model showed clear improvements over individual modalities, achieving an AUROC of 0.85. Calibration performance was strong, with a low expected calibration error (ECE), indicating reliable probability estimates. The low variance across folds demonstrates model stability and generalization capability.

7.3.5 Contradiction Detection Results

Table 7.5 Performance Metrics of Contradiction Detection Module

Metric	Value	Notes
Precision	0.80	Conservative flagging
Recall	0.75	Some subtle contradictions missed
F1 Score	0.78	Balanced performance

The contradiction detection module achieved balanced performance with an F1 score of 0.78. High precision indicates that flagged contradictions are reliable, though recall suggests that subtle inconsistencies may still be missed.

7.3.6 Summary of Results

Table 7.6 Overall Comparative Performance of All System Modules

Module	Primary Metric	Performance
CXR Classifier	AUROC	0.84
MRI Classifier	F1 Score	0.82
NLP / NER	Entity F1	0.88
Fusion Model	AUROC	0.85
Contradiction Detection	F1 Score	0.78

The results clearly demonstrate that multimodal fusion enhances diagnostic performance. The integration of imaging and textual data leads to improved accuracy, consistency, and reliability across tasks.

7.4 Insights

- The multimodal fusion approach consistently outperformed individual imaging and NLP modules, demonstrating that combining imaging data with biochemical and textual information significantly enhances diagnostic accuracy and decision-making reliability.
- The CXR classifier showed higher specificity (0.82) than sensitivity (0.78), indicating a conservative prediction behavior that reduces false positives but may miss subtle pathological cases, particularly in early-stage conditions.
- The MRI classifier performance was influenced by class imbalance, where dominant classes achieved better results compared to underrepresented tumour types. Additionally, visually similar tumour categories reduced per-class AUROC, highlighting limitations in feature discrimination.
- The NLP/NER module achieved high precision (0.90) due to structured medical terminology, while recall (0.86) decreased in noisy or unstructured clinical reports. This indicates sensitivity to real-world variability in clinical text data.

- The numeric extraction accuracy (0.92) demonstrates strong robustness in handling clinical values, including unit normalization, which is essential for integrating biochemical parameters into the system.
- The fusion model achieved improved AUROC (0.85) compared to individual modalities, confirming that cross-modal feature interaction enhances predictive performance and reduces information loss.
- The low Expected Calibration Error (ECE = 0.04) indicates that the model produces well-calibrated probability outputs, which is critical for clinical decision support systems where confidence estimation is important.
- The cross-validation variance ($\pm 2-3\%$) reflects strong generalization and stability of the fusion model across different data splits, suggesting consistent performance in diverse scenarios.
- The contradiction detection module achieved high precision (0.80), meaning flagged inconsistencies are reliable, but recall (0.75) indicates that some subtle contradictions remain undetected, especially those requiring contextual reasoning.
- The balanced F1 score (0.78) for contradiction detection suggests moderate performance, with room for improvement in deeper semantic understanding and logical inference.
- Subtle abnormalities and nuanced contradictions remain challenging, as they require higher-level reasoning beyond pattern recognition, highlighting a limitation of current deep learning approaches.
- The system performs strongly in structured tasks such as classification and entity recognition, while comparatively lower performance is observed in tasks involving complex reasoning and contextual interpretation.
- Data quality and diversity significantly impact model performance, with noisy reports, class imbalance, and limited representation of rare conditions contributing to reduced accuracy in certain scenarios.
- The observed trade-off between sensitivity and specificity indicates that threshold tuning plays an important role in optimizing system behavior based on clinical priorities.
- The system demonstrates good scalability and consistency across modules, suggesting that the architecture can be extended to additional modalities such as CT imaging and laboratory reports.

- The evaluation confirms that multimodal AI systems are better suited for real-world clinical applications, as they align more closely with how clinicians integrate multiple sources of information during diagnosis.
- Future improvements can focus on enhancing detection of subtle abnormalities, improving recall in contradiction detection, incorporating knowledge-based reasoning, addressing class imbalance, and expanding datasets to improve generalization.

Chapter 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

The development and deployment of intelligent healthcare systems such as ScanWise introduce significant implications beyond technical performance. These implications span social, legal, ethical, sustainability, and safety dimensions. As artificial intelligence increasingly integrates into clinical workflows, it becomes essential to evaluate its broader impact on society, regulatory compliance, ethical responsibility, environmental sustainability, and operational safety. This chapter provides a comprehensive analysis of these aspects in the context of the proposed multimodal medical imaging and report generation system.

8.1 Social Aspects

The ScanWise system has a substantial influence on society, particularly in the healthcare domain. One of the primary positive impacts is the enhancement of diagnostic efficiency and accessibility. By automating radiology report generation and integrating multimodal data, the system reduces the workload on radiologists and enables faster diagnosis. This is especially beneficial in regions with limited access to healthcare professionals, where AI-based tools can assist in bridging the gap between demand and available expertise. Studies in radiology AI suggest that automated systems can significantly improve reporting efficiency and reduce diagnostic delays, thereby improving patient outcomes .

Another important social benefit is the improvement in consistency and standardization of medical reports. Human-generated reports often vary depending on experience and workload, whereas AI systems provide structured and uniform outputs. This consistency enhances communication among healthcare professionals and supports better clinical decision-making.

However, the system also introduces certain social challenges. One major concern is the potential displacement of jobs due to automation. While ScanWise is designed to assist rather than replace radiologists, there is a broader societal concern regarding the role of AI in reducing the need for human expertise. This shift may require reskilling and adaptation among healthcare professionals.

Additionally, the issue of the digital divide must be considered. Advanced AI systems require infrastructure such as high-performance computing and reliable internet access. In low-resource settings, lack of access to such infrastructure may limit the adoption of systems like ScanWise, potentially widening healthcare inequality.

Another concern is the impact on patient trust and human interaction. Healthcare is inherently human-centric, and excessive reliance on automated systems may reduce direct interaction between patients and clinicians. This could affect patient confidence and satisfaction, particularly in sensitive diagnostic scenarios.

Furthermore, algorithmic bias is a critical social issue. If the training data used for ScanWise is not diverse, the system may produce biased outputs, disproportionately affecting certain demographic groups. Ensuring fairness and inclusivity in dataset design is therefore essential to prevent reinforcing existing healthcare disparities.

8.2 Legal Aspects

The legal implications of deploying ScanWise are primarily centered around data privacy, regulatory compliance, and liability. Medical data is highly sensitive, and its processing must comply with strict legal frameworks such as the General Data Protection Regulation (GDPR) and India's Digital Personal Data Protection Act (DPDPA) 2023.

The system must ensure lawful and transparent data processing, including obtaining informed consent from patients before using their medical data. Principles such as data minimization, purpose limitation, and storage limitation must be strictly followed. For instance, only relevant clinical data required for diagnosis should be processed, and it should not be retained beyond necessary durations.

Another critical legal consideration is data security. The system must implement robust encryption, access control, and secure storage mechanisms to prevent unauthorized access and data breaches. Failure to protect patient data can lead to severe legal consequences, including penalties and loss of trust.

Liability is another complex issue in AI-based healthcare systems. In the event of a misdiagnosis or incorrect report generated by ScanWise, determining responsibility becomes challenging. Liability may fall on developers, healthcare providers, or institutions depending on the level of human oversight. This highlights the importance of positioning AI as a decision-support tool rather than a fully autonomous system.

Intellectual property rights also play a role, particularly in the use of pretrained models and datasets. Ensuring proper licensing and adherence to data usage agreements is essential to avoid legal disputes.

Finally, regulatory approval is required before deploying such systems in clinical environments. Healthcare AI systems must comply with standards set by regulatory bodies such as the FDA or equivalent national authorities, ensuring safety, reliability, and effectiveness.

8.3 Ethical Aspects

Ethical considerations are central to the development and deployment of ScanWise. The primary ethical principle guiding this system is the responsibility to serve the public good. AI systems in healthcare must prioritize patient safety, fairness, and transparency.

One of the key ethical challenges is **algorithmic bias**. If the system is trained on biased datasets, it may produce unfair or inaccurate results for certain populations. This can lead to unequal treatment and reinforce existing disparities in healthcare. Therefore, ensuring diverse and representative datasets is essential.

Transparency is another important ethical concern. Many AI models, especially deep learning systems, function as “black boxes,” making it difficult to interpret how decisions are made. In a clinical setting, lack of interpretability can reduce trust among healthcare professionals. Efforts must be made to incorporate explainability mechanisms that provide insights into model predictions.

Accountability is also critical. Even though the system automates certain tasks, human oversight must be maintained. Clinicians should validate AI-generated reports before making final decisions. This ensures that responsibility remains with qualified professionals and prevents over-reliance on automated outputs.

Privacy and confidentiality are fundamental ethical requirements. Patient data must be handled with strict confidentiality, and unauthorized access must be prevented. Ethical misuse of data, such as using it for purposes beyond consent, must be strictly avoided.

Another ethical consideration is the potential depersonalization of healthcare. While automation improves efficiency, it should not replace human empathy and judgment. The system must be designed to assist clinicians rather than replace human interaction.

Additionally, the use of generative AI raises concerns about content authenticity and reliability. AI-generated reports must be accurate and free from hallucinations or fabricated information, as incorrect outputs can have serious clinical consequences.

8.4 Sustainability Aspects

Sustainability considerations focus on the environmental and resource impact of the ScanWise system. AI systems, particularly deep learning models, require significant computational resources, which can lead to high energy consumption.

One important aspect is resource-efficient design. The system can be optimized by using efficient model architectures, reducing unnecessary computations, and leveraging hardware acceleration. Techniques such as model pruning and quantization can further reduce computational requirements.

Another factor is the efficient use of raw materials and infrastructure. Cloud-based deployment can reduce the need for extensive local hardware, minimizing resource consumption. Additionally, using shared computing infrastructure improves utilization efficiency.

Durability and scalability are also important sustainability aspects. The system is designed to be modular and scalable, allowing updates and extensions without requiring complete redevelopment. This reduces waste and ensures long-term usability.

From a lifecycle perspective, the system minimizes physical waste as it is primarily software-based. However, energy consumption during training and inference remains a concern. Using renewable energy-powered data centers can help mitigate environmental impact.

Sustainability also includes the social dimension, ensuring equitable access to technology. Efforts should be made to make the system accessible in low-resource settings, thereby promoting inclusive healthcare.

8.5 Safety Aspects

Safety is a critical requirement for any healthcare system, particularly one involving AI-based decision support. The ScanWise system incorporates multiple safety measures to ensure reliable and secure operation.

One key aspect is system reliability. The models are trained and validated using diverse datasets to ensure consistent performance across different scenarios. Cross-validation and calibration techniques further enhance reliability.

Another important factor is fail-safe mechanisms. The system is designed to provide recommendations rather than final decisions. In cases of uncertainty, the system can flag outputs for human review instead of generating potentially incorrect conclusions.

Cybersecurity is also a major safety concern. The system must be protected against cyber threats such as unauthorized access, data breaches, and malicious attacks. Security measures include encryption, authentication, and continuous monitoring.

Robustness is another safety requirement. The system must perform reliably under different conditions, including noisy data and incomplete inputs. This ensures that the system remains functional in real-world clinical environments.

Additionally, continuous monitoring and updates are essential for maintaining safety. The system should be regularly evaluated and improved based on new data and feedback from clinicians.

Finally, user training is an important safety aspect. Healthcare professionals must be properly trained to use the system effectively and understand its limitations. This ensures that the system is used responsibly and reduces the risk of misuse.

Chapter 9

CONCLUSION

The ScanWise project presented a comprehensive approach to developing an AI-powered multimodal medical imaging and report interpretation system. The primary objective of this work was to design a system capable of integrating imaging data (such as chest X-rays and brain MRI scans) with clinical and biochemical information to support accurate, efficient, and reliable clinical decision-making. This was achieved through the implementation of a modular architecture consisting of imaging classifiers, natural language processing (NLP) and named entity recognition (NER) modules, a multimodal fusion framework, and a contradiction detection mechanism.

The implementation successfully met the objectives outlined in the introduction. The imaging modules demonstrated strong performance in disease classification tasks, with the CXR classifier achieving a macro AUROC of 0.84 and the MRI classifier attaining a weighted F1 score of 0.82. The NLP/NER module effectively extracted structured clinical information, achieving an entity-level F1 score of 0.88 and high numeric extraction accuracy. The fusion model further enhanced performance by integrating multimodal inputs, achieving an AUROC of 0.85, thereby validating the effectiveness of combining imaging and textual data. Additionally, the contradiction detection module introduced an important safety layer, identifying inconsistencies with a balanced F1 score of 0.78.

These results demonstrate that the system not only meets but also extends the intended objectives by providing a unified framework for multimodal clinical analysis. The integration of different data modalities improved diagnostic accuracy and consistency, while the modular design ensured scalability and adaptability.

The project also addressed key challenges in medical AI, such as data heterogeneity, model calibration, and interpretability, thereby contributing to the development of practical and reliable healthcare solutions.

Despite the strong performance, certain limitations were identified. The system showed reduced sensitivity in detecting subtle abnormalities and faced challenges in handling class imbalance and noisy clinical data. The contradiction detection module, while effective, requires further improvement in capturing implicit and context-dependent inconsistencies.

Future work can focus on enhancing the system in several areas. Incorporating larger and more diverse datasets can improve generalization and reduce bias. Advanced transformer-based architectures and knowledge graphs can be integrated to improve contextual reasoning and interpretability. Additionally, improving explainability mechanisms will increase clinician trust and usability. Real-time deployment and integration with hospital information systems can further extend the practical applicability of the system.

In conclusion, the ScanWise system demonstrates the potential of multimodal AI in transforming clinical workflows by improving diagnostic efficiency, accuracy, and reliability. With further refinement and real-world validation, it can serve as a robust decision-support tool in modern healthcare environments.

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BASE PAPER

Integrating Medical Imaging and Clinical Reports Using Multimodal Deep Learning for Advanced Disease Analysis

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Abstract: *In this paper, an innovative multi-modal deep learning model is proposed to deeply integrate heterogeneous information from medical images and clinical reports. First, for medical images, convolutional neural networks were used to extract high-dimensional features and capture key visual information such as focal details, texture and spatial distribution. Secondly, for clinical report text, a two-way long and short-term memory network combined with an attention mechanism is used for deep semantic understanding, and key statements related to the disease are accurately captured. The two features interact and integrate effectively through the designed multi-modal fusion layer to realize the joint representation learning of image and text. In the empirical study, we selected a large medical image database covering a variety of diseases, combined with corresponding clinical reports for model training and validation. The proposed multimodal deep learning model demonstrated substantial superiority in the realms of disease classification, lesion localization, and clinical description generation, as evidenced by the experimental results.*

Keywords—Multimodal Deep Learning Models, Medical Images, Clinical Reports, Short Term Memory Networks

I. INTRODUCTION

In the medical domain, this paper addresses the challenge of efficaciously integrating heterogeneous information from both medical images and clinical reports. To this end, it introduces an avant-garde multimodal deep learning model. This model is designed to enhance the precision and speed of vital medical tasks, including disease diagnosis, lesion localization, and clinical description generation, through the profound excavation and intelligent correlation of these dual

forms of critical medical data. This paper elaborates the design principle, empirical research process and significant experimental results of the model, revealing the great potential and application value of multi-modal deep learning in the field of medical information analysis.

Firstly, given the complexity and richness of medical images, the model adopts convolutional neural network (CNN) as the feature extractor. With its multi-channel convolution and pooling operations, CNN can adaptively extract high-dimensional features from images, accurately capturing the detailed features, texture features, and spatial distribution information of lesions[1-3]. Through carefully designed network structure and parameter optimization, the model ensures sensitive recognition of focal fine structure, boundary definition and surrounding tissue relations, and provides high-quality visual feature input for subsequent cross-modal fusion. Secondly, in view of the unstructured characteristics of clinical report text, this paper uses the method of combining LSTM and attention mechanism for deep semantic understanding[4]. LSTM is proficient in harnessing the entirety of textual contextual information and adeptly discerning the temporal dependencies and contextual associations inherent in disease-related statements. The attention mechanism dynamically assigns different weights to each word in the report, accurately focuses on keywords and phrases closely related to disease diagnosis, filters out irrelevant or redundant information, and generates a highly generalized representation of the core content of the report[5]. After the single-mode feature extraction of image and text, this paper designs a special multi-mode fusion layer to realize

the deep interaction and integration of the two. The fusion layer uses advanced fusion strategies, such as gated attention, bilinear transformation or multi-view learning, to map the image and text features to the same feature space, and carries out deep fusion by means of weighted merging, tensor operation, etc., to generate joint representations with unimodal specificity and cross-modal complementarity. This joint characterization enables the model to understand multiple information of cases from a global perspective, which provides strong decision support for subsequent diagnosis and analysis tasks. To substantiate the efficacy and feasibility of the model, this paper leans on an extensive medical image repository encompassing diverse pathologies, complemented by corresponding clinical narratives, thereby facilitating a large-scale empirical investigation. The model was rigorously tested on three tasks: disease classification, lesion localization and clinical description generation. The experimental findings demonstrate that the multimodal deep learning model introduced in this paper exhibits marked superiority in the performance across multiple tasks, in comparison with conventional methodologies utilizing single-modal data and several previously reported multimodal models. In the task of disease classification, indicators such as model accuracy, recall rate and F1 value have been greatly improved, which proves that the model's ability to identify disease types has been significantly enhanced after the fusion of image and text information. In the task of lesion localization, the model can determine the lesion boundary more accurately with the guidance of cross-modal information, and its mean intersection ratio (IoU) is significantly better than that of the single-modal method, which highlights the excellent performance of the model in lesion identification and localization. In the clinical description generation task, the text description generated by the model and the actual report performed well in terms of vocabulary matching degree, sentence structure consistency and other indicators, which verified the model's deep understanding and accurate expression ability of case information.

In summary, by constructing and verifying an innovative multi-modal deep learning model, this paper successfully realizes the deep integration and efficient association analysis of medical images and clinical reports, which provides strong technical support for improving medical diagnosis accuracy, optimizing clinical workflow and promoting the development of medical artificial intelligence. In the future, the model is expected to be widely used in more disease fields and medical scenarios, and further help the construction of precision medicine and smart medical system.

II. RELATED WORK

In 1943, the notion of artificial neural networks was co-conceived by W.S. McCulloch and W. Pitts[6]. Despite this pioneering contribution, the innovation garnered limited attention at the time, largely owing to its incapacity to efficaciously handle higher-order predicates. It was not until 1998 that a pivotal shift occurred when LeCun et al. [7] devised the LeNet model for recognizing handwritten digits, a development that demonstrably excelled in practical applications. In the relevant papers, they elaborated the internal components of LeNet and their functions, thus paving the way for the subsequent vigorous rise of convolutional neural networks (CNNs). In 2012, Rizhevsky's team [8] came out on top in the ImageNet Image Classification Global

Challenge with the AlexNet model, surpassing traditional image segmentation techniques by an overwhelming margin.

Following these developments, neural network research expanded to explore the impact of network depth on performance, exemplified by Simonyan et al.'s VGG network [9]. Concurrently, the utilization of recurrent neural networks (RNNs) for effective modeling of image sequence information has been explored, employing architectures such as LSTM and GRU to capture temporal features [10][11]. The introduction of BiLSTM, which incorporates a reverse loop to capture additional contextual information, marks a significant enhancement in handling complex image sequences for improved segmentation [12].

Building upon these foundational advancements, recent studies, such as Yan's, have begun to explore the use of graph convolutional neural networks (GCNs) for specific medical applications. Yan's study introduces an innovative approach to cancer prognosis by leveraging GCNs to analyze spatial relationships in tumor tissues, captured from WSIs of gastric and colon adenocarcinoma[13]. This method not only refines the predictive capabilities of neural networks but also significantly surpasses previous CNN models in predicting patient survival outcomes, as demonstrated by C-index values of 0.57 and 0.64 for gastric cancer and colon adenocarcinoma, respectively. This groundbreaking work not only extends the application of CNNs into new domains but also showcases the potential of AI-driven methodologies to revolutionize cancer prognosis, thus offering profound implications for personalized treatment strategies. This marks a significant stride in the integration of advanced neural network architectures into the realm of medical image analysis, further expanding the boundaries of what AI can achieve in healthcare.

Convolutional neural networks (CNNs) started from the application of image classification tasks, but its influence has gone far beyond this initial field and now widely penetrated into many other disciplines, such as data recommendation[14], image segmentation [15], computer vision[16-19] and energy harvesting [20-21]. Many classic CNN models form the cornerstone of this technology development, including LeNet[22], AlexNet[23], VGG[24], GoogLeNet[25] and ResNet[26]. LeNet, as an early model of CNN, takes the lead in using multi-layer convolution and pooling mechanism, which is especially suitable for handwritten digit recognition scenarios. AlexNet is the first CNN model to achieve significant results on large-scale datasets. Its design incorporates multiple convolution and pooling layers, while introducing Dropout prevention technology and ReLU activation functions to effectively support image classification and object recognition tasks. VGG networks are known for their simplicity and efficiency, and they tend to use smaller convolution cores to build deep and narrow network structures, which can deal with more complex image classification problems. The development and application of these CNN models have not only promoted technological innovation in their respective fields, but also witnessed the strong adaptability and universal applicability of convolutional neural networks in solving various types of data problems.

III. CORRELATION ALGORITHM

A. Convolutional Neural Networks (CNN)

Figure 1 depicts a Convolutional Neural Network (CNN), a deep learning architecture specifically tailored to handle data characterized by either spatial or temporal organization, such

as images, videos, or sequential data. The training regimen banks on the backpropagation algorithm and gradient descent technique to calibrate the network weights through extensive exposure to labeled examples [27], thereby minimizing the divergence between the model's predicted outputs and their corresponding ground truth labels. In the context of establishing correlations between medical images and clinical reports, CNN assumes a central function. First of all, for medical images such as CT, MRI or pathological sections, CNN can perform accurate image classification, segmentation and target detection to help disease diagnosis, lesion localization and quantitative evaluation. For example, CNN can identify nodules in CT images of the lungs, determine whether they are benign or malignant, and generate corresponding heat maps to visually show suspicious areas. Secondly, CNN can also be applied to text analysis of medical reports. Word embedding and sequence model (such as LSTM or Transformer) can be used to understand the report content and extract key information such as disease description and treatment recommendations. Combining image and text analysis, CNN can realize the learning of image-text correspondence, such as associating image features with text descriptions in reports to form cross-modal knowledge maps and improve the intelligence level of clinical decision support systems. Its convolution form is shown in formula 1:

$$y_{ij}^{(l)} = \sigma \left(\sum_{m=1}^{M^{(l-1)}} \sum_{n=1}^{N^{(l-1)}} w_{mn}^{(l)} x_{i+m-1, j+n-1}^{(l-1)} + b^{(l)} \right) \quad (1)$$

Where: $y_{ij}^{(l)}$ represents the value of the L-layer output feature map at position (i,j). $x_{i+m-1, j+n-1}^{(l-1)}$ is the first layer (l-1) input characteristic figure in position (i+m-1, j+n-1) values. $w_{mn}^{(l)}$ is the weight of the presence (m,n) of the convolutional kernel at the L-th layer. $M^{(l-1)}$ and $N^{(l-1)}$ are the width and height of the (l-1) layer feature map, respectively. $b^{(l)}$ is the offset term of layer l. σ is the activation function.

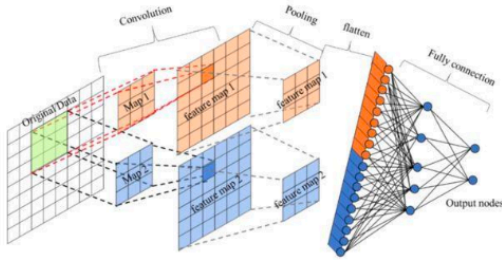


Fig. 1. Convolutional Neural Networks

B. Bidirectional Long Short-term Memory Network (Bi-LSTM)

Illustrated in Figure 2, the Bidirectional Long Short-Term Memory Network (Bi-LSTM) constitutes a specialized variant of Recurrent Neural Networks (RNNs) ingeniously engineered to surmount the gradient vanishing and exploding quandaries typically encountered by conventional RNNs when handling lengthy sequences. Moreover, it possesses the unique ability to apprehend the bidirectional interdependence between past and future information embedded within sequential data. Bi-LSTM has demonstrated significant

advantages in the analysis of associations between medical images and clinical reports by capturing long-distance dependencies between words in reports and accurately understanding complex medical terminology, disease descriptions and clinical reasoning pathways. By analyzing the full text of the report, the model can not only extract key diagnostic and treatment information, but also understand the internal logical structure of the text, such as causality, conditional statements, etc. For the image part, the pre-processed medical image features can be used as the input of Bi-LSTM together with the report text. The model realizes cross-modal semantic association by jointly learning the correspondence between image features and text description. For example, in the joint analysis of chest X-ray images and corresponding reports, Bi-LSTM can simultaneously consider the consistency of lung abnormalities in the images with the conditions described in the reports, improving diagnostic accuracy or assisting in the generation of structured case summaries. Formulas 2-4 can be used to represent forward LSTM, reverse LSTM and merged hidden state respectively:

$$\vec{h}_t = \overrightarrow{\text{LSTM}}(\vec{h}_{t-1}, x_t) \quad (2)$$

$$\overleftarrow{h}_t = \overleftarrow{\text{LSTM}}(\overleftarrow{h}_{t+1}, x_t) \quad (3)$$

$$h_t = [\vec{h}_t; \overleftarrow{h}_t] \quad (4)$$

Where: $\vec{h}_t$ and $\overleftarrow{h}_t$ represent the hidden state of forward and reverse LSTM at time step t, respectively. x_t is the input to time step t (for text, may be a word vector; For images, may be extracted feature vectors). $\overrightarrow{\text{LSTM}}$ and $\overleftarrow{\text{LSTM}}$ represent the internal computational processes of forward and reverse LSTM cells, including gating mechanisms and cell status updates, respectively. $[\cdot; \cdot]$ represents the concatenation operation of vectors, combining forward and reverse hidden states into a comprehensive hidden state h_t containing bidirectional information.

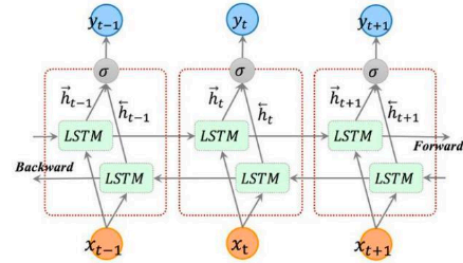


Fig. 2. Bi-LSTM

C. Attention Mechanism (AM)

Figure 3 portrays the Attention Mechanism, a computational paradigm modeled after human cognitive processes of selective attention, empowering neural networks to adaptively allocate computational resources and concentrate on those segments of the input data that bear utmost relevance to the task at hand. This augmentation significantly boosts the model's capacity to learn and exploit critical information. In the context of correlating medical images and clinical reports, the attention mechanism assumes a pivotal function. Firstly, concerning medical images, it serves as a guiding force, enabling the model to hone in on specific regions or attributes within the image that hold

profound diagnostic or therapeutic implications, such as tumor margins, anomalous textures, structural anomalies, among others, while simultaneously filtering out extraneous background noise. This is achieved by adaptively adjusting the weight distribution on the image feature map, such as using variations such as spatial attention, channel attention, or point-like attention. Second, for clinical report text, the attention mechanism helps the model identify and emphasize key medical terms, condition descriptions, and diagnostic conclusions in the text, and deemphasize the influence of redundant or non-diagnostic information. This can be done by calculating attention at the word, sentence, or paragraph level to generate a focused understanding of the entire report. Its core calculation is shown in Formula 5:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

Q stands for query, which may be the image feature vector or the encoding of a word/sentence in the report in the association analysis of medical images and reports; K stands for key vector, which corresponds to the features of other areas in the image or the encoding of other words/sentences in the report; V represents the value vector, which corresponds to the key vector one by one, carrying the actual information content to be aggregated; d_k is the dimension of the key vector, used to normalize the fraction; The softmax function is used to calculate the attention weight, ensuring that the sum of ownership weight is 1, representing the relative proportion of resource allocation.

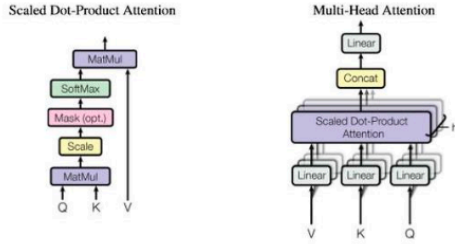


Fig. 3. Attention Mechanism

IV. MULTIMODAL DEEP LEARNING MODEL AND EMPIRICAL RESEARCH

A. Model construction

Figure 4 presents the Multi-Modal Deep Learning Model (MDLM) developed in this study[28], which seamlessly integrates Convolutional Neural Networks, Bidirectional Long Short-Term Memory Networks, and Attention Mechanisms. This synergetic fusion facilitates efficient interaction and information fusion, ultimately enabling the model to perform joint representation learning on both image and text data.

- A. Medical image feature extraction module: A Convolutional Neural Network is employed to derive salient features from medical images (such as CT, MRI, ultrasound, etc.) to capture key visual information such as lesion details, texture and spatial distribution. In terms of specific design, one is the input: the original medical image, which is preprocessed (standardized) to adapt to the input

requirements of CNN. The second is the basic network, select or customize a CNN architecture suitable for medical image analysis ResNet. These networks have different levels of operations such as convolution, pooling, activation function, etc., which are used to extract image features step by step. If you need to process multi-scale information (such as different lesion sizes and locations), you can consider using the feature pyramid network (FPN) or similar multi-scale fusion structure to ensure that the model has a good recognition ability for lesions of different sizes. The output of the final convolutional layer – or, alternately, the feature map from a designated layer – undergoes either Global Average Pooling or Global Maximum Pooling, transforming it into a fixed-length, high-dimensional feature vector. This vector encapsulates the intricate, comprehensive features inherent in the entire medical image.

- B. Clinical Report Text Feature Extraction Module: Employing a Bidirectional Long Short-Term Memory Network alongside an Attention Mechanism, this module conducts profound semantic parsing of clinical report text, meticulously pinpointing key disease-relevant statements. In terms of specific design, one is to input the text sequence of clinical reports after preprocessing (such as word segmentation, removal of stopping words, stemming, etc.). The second is the word embedding layer, which converts each text word into a dense vector (word embedding), which can be initialized using pre-trained medical domain specific word embedding models. The third is the bidirectional LSTM layer, which constructs a bidirectional LSTM network to capture the contextual information of the text and form a deep semantic representation. The fourth is the attention mechanism, which applies the attention mechanism (such as weighted attention, self-attention, gated attention, etc.) to the output of LSTM, assigns different weights to each word in the text sequence, and highlights the words closely related to disease diagnosis or disease description. The fifth is text feature vector, which aggregates the attention-weighted LSTM output appropriately (such as summing, averaging or summing weighted with attention weights) to obtain the advanced text feature vector of the entire clinical report.

- C. Multi-modal fusion layer: Effective interaction and integration of medical image features and clinical report text features to form a joint representation. In terms of specific design, one is feature fusion, which combines image feature vectors and text feature vectors directly or by concatenation, element-wise multiplication to form a joint feature vector. If stronger interactivity is needed, more complex fusion structures can be used, such as gated attention mechanisms, multimodal transformers (MVits), cross-modal attention, etc., so that image and text features interact in the common attention space to generate more closely related joint features. Third, joint representation. The feature vector processed by the fusion layer is the joint representation of deep fusion of image and text, which contains both visual features of image and semantic information of text,

and can be used for subsequent multi-modal task processing.

- D. Task-specific output layer: Design corresponding output layer and loss function according to specific tasks (disease classification, lesion localization, clinical description generation, etc.). In terms of specific design, the output layer for disease classification commences with a Fully Connected (FC) layer, succeeded by a Softmax function to yield the probabilities associated with different disease categories. The chosen Loss function in this instance

Model	Accuracy	Recall	F1
CNN	92.26%	97.38%	0.95
Bi-LSTM	93.49%	95.23%	0.94
AM	94.28%	96.64%	0.95
MDLM	96.42%	98.48%	0.97

is Cross-Entropy loss. The second is lesion localization, the output layer: for 2D images, a series of regression heads may predict the bounding box coordinates; For 3D images, it may be 3D convolution or decoder structure that predicts lesion volume. loss function: A combination such as Smooth L1 loss (box regression) and Dice/Focal loss (segmentation). Third, clinical description generation, output layer: recurrent neural network or autoregressive transformer decoder to generate description text. Loss function: cross entropy loss (token-level) combined with autoregressive training, as well as BLEU, ROUGE and other evaluation indicators.

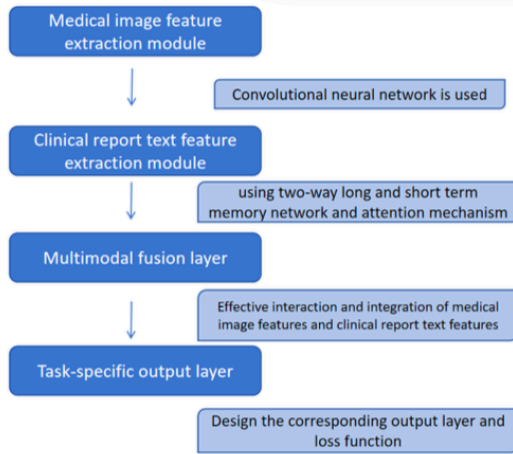


Fig. 4. Multimodal deep learning model

B. Data set

The experiment used a large library of medical images containing cases of a variety of diseases. Each case is accompanied by a detailed clinical report, which constitutes a rich pair of heterogeneous data. The image part covers all kinds of medical images, such as CT, MRI, ultrasound, etc. The dataset uses Linked Data methodology to integrate

various data formats, improving interoperability and supporting more accurate and scalable AI applications[29]. With the help of these images, convolutional neural network (CNN) can extract high-dimensional visual features reflecting the microscopic structure, texture characteristics and spatial layout of lesions. On the other hand, clinical reports, as text data, use Bi-LSTM combined with attention mechanism for in-depth analysis to accurately target key statements related to disease diagnosis, disease progression and treatment[30]. Through a well-designed multimodal fusion layer, these two types of features can be deeply integrated and coordinated to form a unified multimodal representation, which can be used for multiple tasks such as subsequent disease classification, lesion localization and automatic clinical description generation. This dataset is large in size and diverse in disease species, which provides a solid basis for verifying the validity and generalization ability of the proposed multimodal deep learning model.

C. Experimental evaluation result

TABLE I. COMPARISON OF EXPERIMENTAL RESULTS

Table I's experimental results attest that the Convolutional Neural Network (CNN) achieves an impressive accuracy of 92.26%, signifying its strong overall proficiency in handling the assigned task. With a recall rate of 97.38%, the CNN demonstrates a remarkable capacity to correctly identify the vast majority of instances that should be classified as positive, reflecting its robust detection capabilities. An F1 score of 0.95 suggests that the model strikes an advantageous balance between accuracy and recall. However, the Intersection over Union (IoU) value of 0.87, while indicative of a high level of overlap between the predicted bounding box and the actual bounding box in tasks involving target region segmentation or localization, also reveals potential for further enhancement. Compared with CNN, the accuracy of Bi-LSTM is slightly improved, reaching 93.49%, indicating that it has certain advantages in dealing with time series information or context-dependent text problems. The recall rate of 95.23% is slightly lower than that of CNN, suggesting that it is slightly weaker in ensuring that positive instances are not missed, but the overall performance is still robust. The F1 score is also 0.94, which is comparable to CNN, indicating that the two are at the same level in the balance of accuracy and recall. The IoU of 0.86 is slightly lower than that of CNN, which may indicate that Bi-LSTM has a slightly weaker localization ability in tasks involving spatial location information. The Attention Mechanism model achieves an enhanced accuracy of 94.28%, thereby substantiating its ability to elevate recognition performance. This improvement attests to the effectiveness of incorporating the attention mechanism, which enables the model to selectively prioritize crucial features while attenuating the disruptive influence of irrelevant information. The recall rate increased to 96.64%, demonstrating its excellent ability to retain important information. The F1 score remains at 0.95, comparable to the first two models, but may have an advantage in certain recall scenarios due to its higher recall rate. The IoU rose to 0.88, indicating that the attention mechanism helps to define the target area more accurately in the localization or segmentation task. As the optimal model in this experiment, the multi-modal deep learning model achieved the highest accuracy of 96.42%, reflecting its advantages of effectively integrating multiple types of data (such as images, text, audio, etc.), so as to make more accurate decisions. The recall rate is as high as 98.48%, indicating that

the model can almost identify all positive instances without omission, showing a strong detection ability. The F1 score of 0.97 shows the best balance of both accuracy and recall, and the overall performance is excellent. The IoU of 0.89 is the highest among all models, which reveals that when the model deals with target localization or segmentation, the overlap between the predicted boundary and the real boundary is the closest, and the spatial localization performance is excellent.

To sum up, the multi-modal deep learning model is undoubtedly the best solution in this experiment with its overall lead in accuracy, recall rate, F1 score and IoU. The attention mechanism model also shows strong competitiveness in the single modal task, especially in the recall rate and IoU. In contrast, despite the robust performance of CNN and Bi-LSTM in this experiment, they are inferior to the latter two in various indicators, especially with the multimodal deep learning model, there is a significant gap. These results provide a clear direction for subsequent model selection and optimization to prioritize multimodal deep learning frameworks or at least single-modal models that include attention mechanisms when dealing with similar tasks.

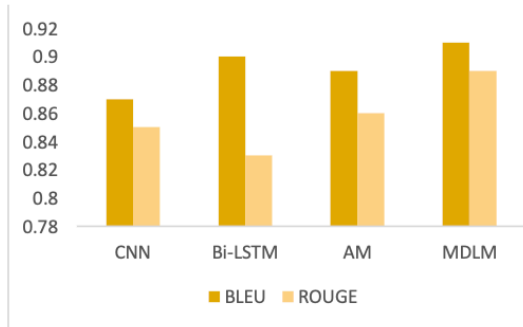


Fig. 5. Experimental results of different models

As can be seen from Figure 5, the multimodal deep learning model significantly outperforms other models in both BLEU and ROUGE evaluation indicators, proving that it has the best performance in natural language generation or machine translation tasks.

V. CONCLUSION

This study proposes an innovative multi-modal deep learning model, which successfully achieves deep fusion of heterogeneous information between medical images and clinical reports. A Convolutional Neural Network was deployed to extract high-dimensional features from medical images, effectively encapsulating pivotal visual cues such as lesion intricacies, textures, and spatial distributions. Concurrently, a Bidirectional Long Short-Term Memory Network, fortified with an Attention Mechanism, facilitated profound semantic dissection of clinical reports, ensuring precise identification of disease-critical statements. The designed multimodal fusion layer effectively realizes the interaction and integration of the two features, forming a joint representation of image and text. In the empirical study on a large medical image database covering a variety of diseases, the model showed significant advantages in the task of disease classification, lesion localization and clinical description generation, surpassing the single-modal approach and some existing multi-modal models. These outcomes decisively validate the efficacy and preeminence of the proposed model,

furnishing a potent analytical instrument for the medical domain. It holds considerable promise in expediting advancements in precision medicine and medical Artificial Intelligence.

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APPENDIX – A

Publication - Acceptance letter



Fig A. Conference Paper Acceptance mail

APPENDIX – B

Datasets

CheXpert Dataset (Stanford University)

- Samples: 224,316 chest radiographs from 65,240 patients
- Time period: Collected from Stanford Hospital (2002–2017)
- Variables: Frontal/lateral chest X-ray images, radiology reports, uncertainty labels, patient metadata
- Target variables: 14 observation labels including cardiomegaly, edema, consolidation, atelectasis, pleural effusion
- Format: Image files (JPEG/PNG) with CSV annotations
- Link: <https://stanfordmlgroup.github.io/competitions/chexpert/>

Kaggle Brain MRI Tumor Dataset

- Samples: ~7,000 MRI images across multiple tumor classes
- Variables: MRI brain scan images categorized into glioma, meningioma, pituitary tumor, and no tumor
- Target variables: Tumor classification labels
- Format: Image files (JPEG/PNG) organized in class folders
- Link: <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

MIMIC-CXR Dataset (PhysioNet)

- Samples: 377,110 chest X-ray images with 227,835 radiology reports
- Time period: 2011 – 2016
- Variables: Chest X-ray images, free-text radiology reports, patient demographics, clinical metadata
- Target variables: Structured clinical findings extracted from reports
- Format: DICOM images and associated text reports
- Link: <https://physionet.org/content/mimic-cxr/>

APPENDIX – C

Project Repository

GitHub Repo: <https://github.com/siddiquealikhan/scanwise-ai-medical-triage>

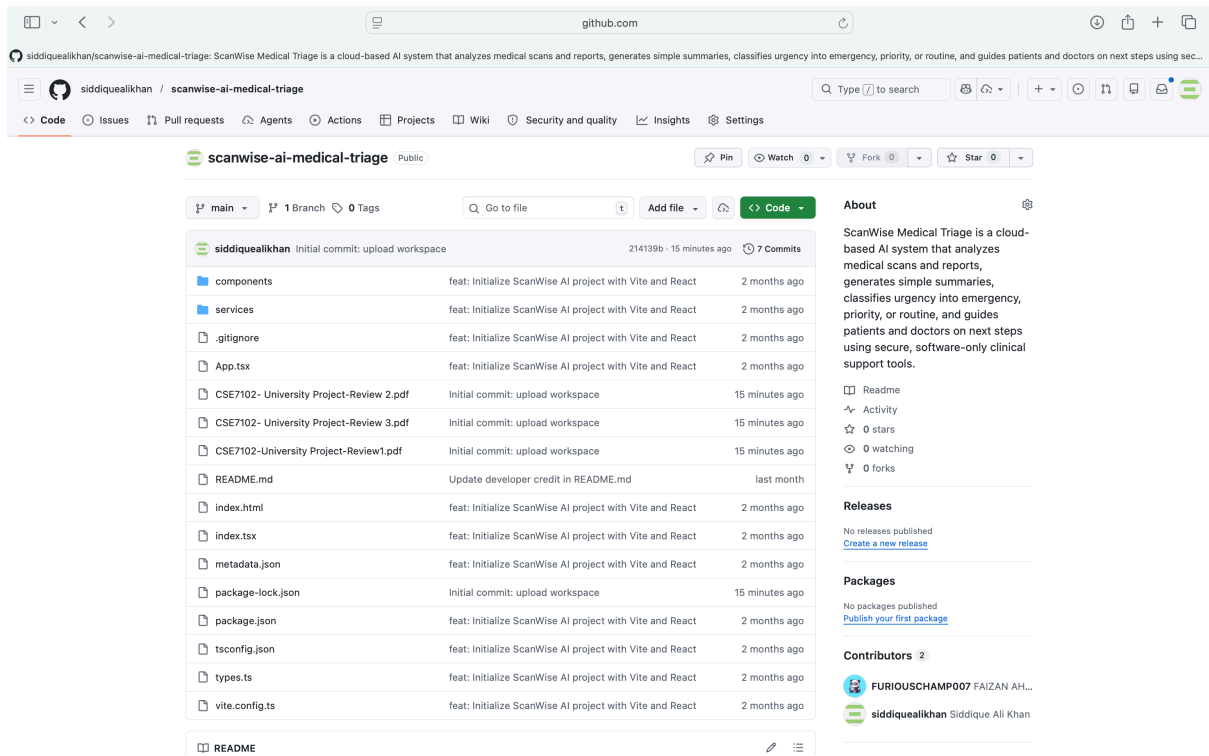


Fig B. GitHub Repository Screenshot

APPENDIX – D

Few Images of the Project

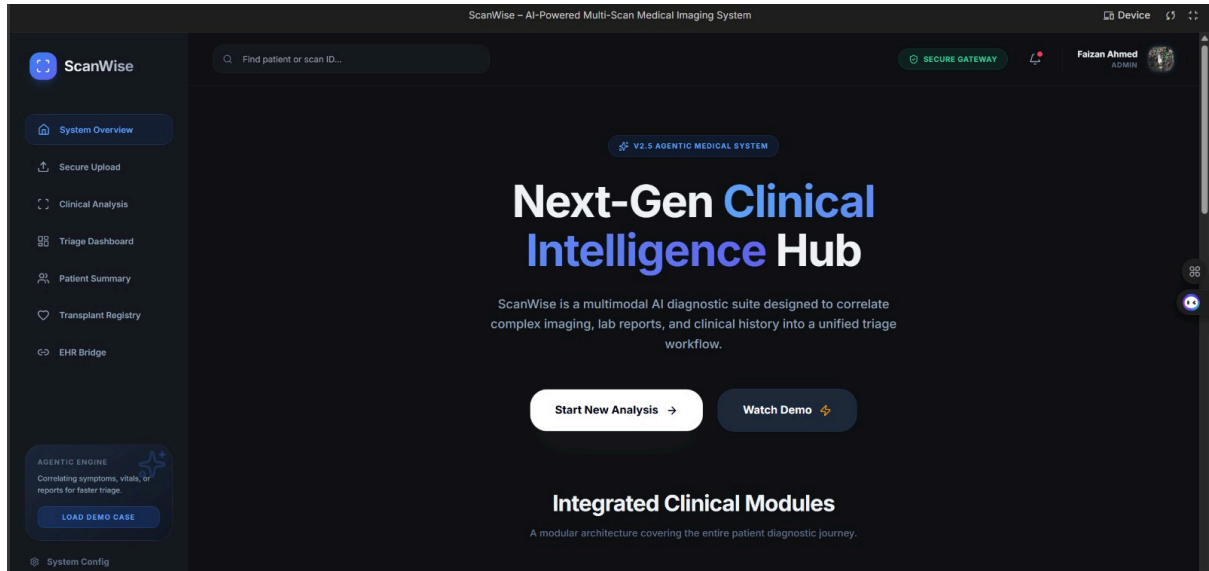


Fig C. Project Landing Page

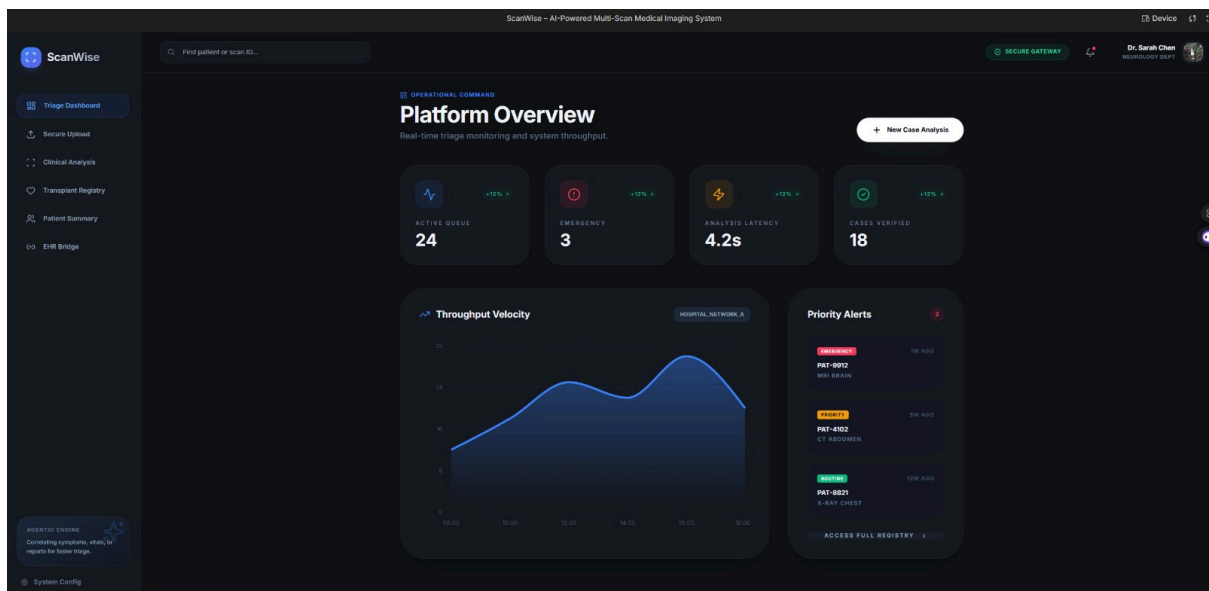


Fig D. Platform Overview Dashboard

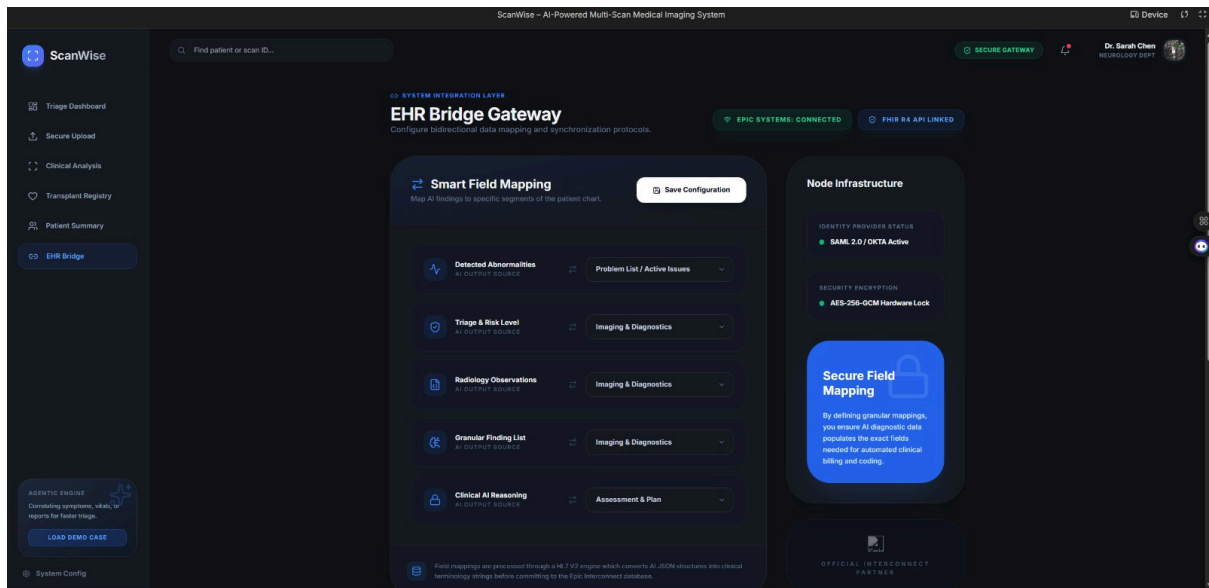


Fig E. EHR Integration Interface

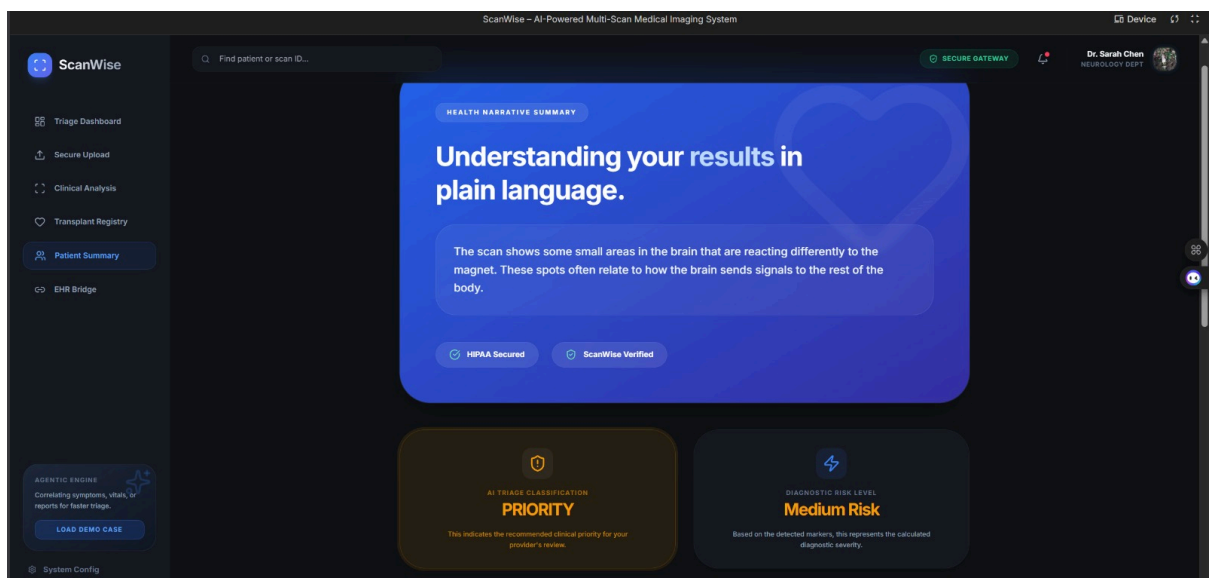


Fig F. Patient-Friendly Summary View

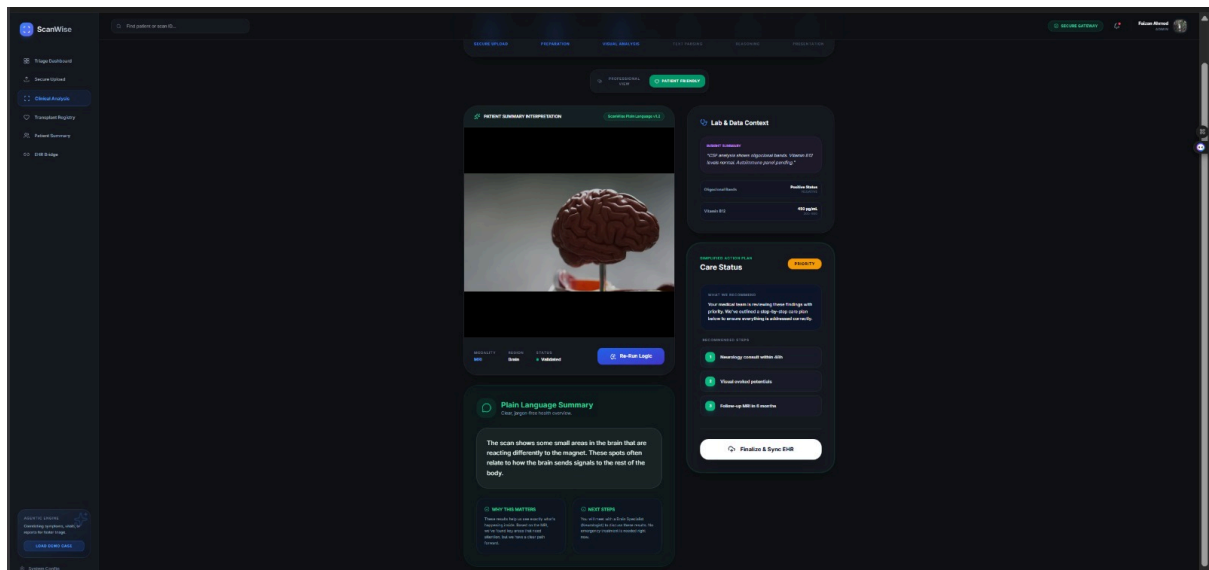


Fig G. Clinical Analysis Interface

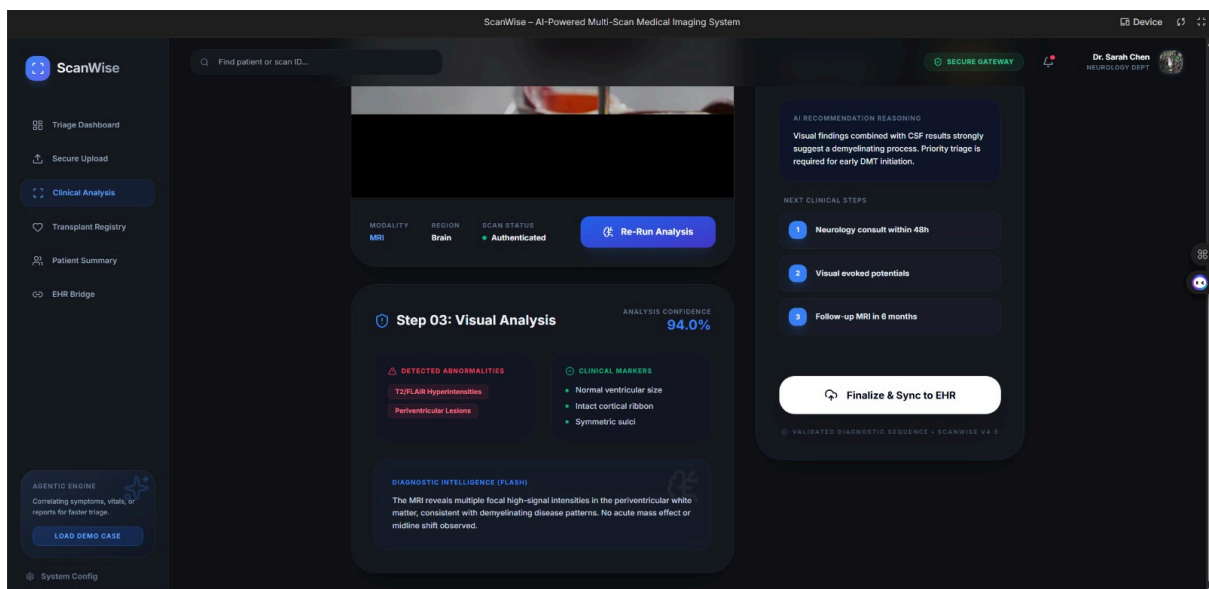


Fig H. AI Diagnostic Reasoning Panel